Part 2: CSS

Introduction

- CSS = Cascading Style Sheets
  - A styling language
  - Current version: CSS3

- Cascading: the procedure that determines which style will apply to a certain section, if you have more than one style rule
- Style: how you want a certain part of your page to look
- Sheets: sets of rules for determining how the webpage will look

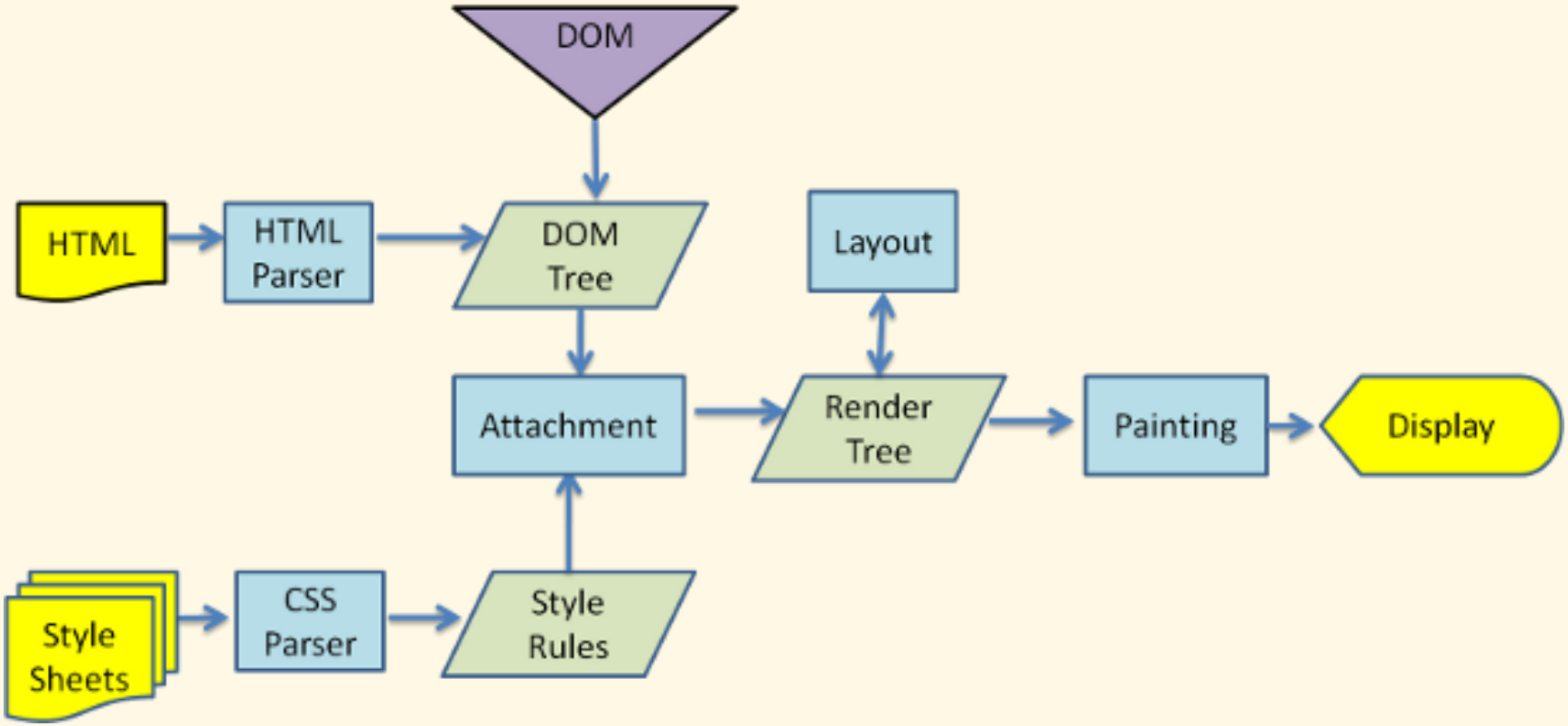
Principle

- A style rule has 2 parts:
  - Selector: the HTML elements you want to add style to
  - Declaration: the statement of style for that element, made up of property and value
- Ex:
  - ```
    selector {
      property1: value1;
      property2: value2;
      //...
    }
    ```
- The selector can either be a grouping of elements, an identifier, class, or single HTML element
    ```

Attaching a Style Sheet

- Attach a style sheet to a HTML page by adding the code to the `<head>` section. There are 4 ways to do.
- Inline Style Sheet: Used directly within HTML tags
  - `<p style="color: red">Some Text</p>`
- Internal Style Sheet: Best used to control styling on one page
  - ```
    <style type="text/css">
    h1 {color: red}
    </style>
    ```
- External Style Sheet: Best used to control styling on multiple pages
  - ```
    <link rel="stylesheet" type="text/css"
    media="all" href="css/styles.css" />
    ```
- Inside another CSS file:
  - `@import url(coolblue.css);`

Rendering Engine Flow



Selectors

Element/Tag Selector

- Specify the style(s) for a single HTML element type
  - ```
    body {
      margin: 0;
      padding: 0;
      border-top: 1px solid #ff0;
    }
    ```
- Grouping elements: allows you to specify a single style for multiple elements at one time
  - ```
    h1, h2, h3, h4, h5, h6 {
      font-family: "Trebuchet MS", sans-serif;
    }
    ```

Identifier (ID) Selector

- Specify the style(s) for element identified by its ID
  - `<p id="first-intro">Introduction text</p>`
  - ```
    #first-intro {
      border-bottom: 2px dashed #fff;
    }
    ```
    ```

Class Selector

- Specify the style(s) for element(s) of user-defined classes
  - `<p class="intro">Introduction text</p>`
  - ```
    .intro {
      font: 12px verdana, sans-serif;
      margin: 10px;
    }
    ```
- An element can use multiple classes:
  - `<p class="intro primary animated">…</p>`
- Identifier or class?
  - An identifier is specified only once on a page and has a higher inheritance specificity than a class
  - A class is reusable as many times as needed in a page
  - Use identifiers for main sections and sub-sections of your document
    ```

Element with Class or ID Selector

- Specify the style(s) for elements of a type, and identified by a class or ID at the same time
  - `p.intro { }`
  - `p#first-intro { }`
  - `p.intro#first-intro { }`
  - `p.intro.primary { }`
  - `p.intro.primary#first-intro { }`
  - `.intro#first-intro { }`

Child Selector

- Specify the style(s) for child element(s) of another
  - ```
    .intro > h1 {
      font-size: 2rem;
    }
    ```
  - ```
    #navigation > p {
      line-height: 20px;
    }
    ```

Descendant Selector

- Specify the style(s) for descendant element(s) of another
  - ```
    body h1 {
      font-size: 2rem;
    }
    ```
  - ```
    #navigation p {
      line-height: 20px;
    }
    ```

Adjacent Sibling Selector

- Specify the style(s) for element that is directly after another
  - ```
    p.intro + h1 {
      font-size: 2rem;
    }
    ```
  - ```
    #navigation + p {
      line-height: 20px;
    }
    ```

General Sibling Selector

- Specify the style(s) for sibling element(s) of another
  - ```
    p.intro ~ h1 {
      font-size: 2rem;
    }
    ```
  - ```
    #navigation ~ p {
      line-height: 20px;
    }
    ```

Attribute Selector

- Specify the style(s) for element(s) with the specified attribute and value
  - Elements with `target` attribute
    - `a[target] { }`
  - Elements whose `target` attribute value is "`_blank`"
    - `a[target="_blank"] { }`
  - Elements whose `title` attribute value contains "`flower`"
    - `a[title~="flower"] { }` (value needs be whole word)
    - `a[title*="flower"] { }` (value doesn't need to be whole word)
  - Elements whose `title` attribute value begins with "`flower`"
    - `a[title^="flower"] { }`
  - Elements whose `title` attribute value ends with "`flower`"
    - `a[title$="flower"] { }`

Universal Selector

- Specify the style(s) for all elements
  - ```
    * {
      color: gray;
    }
    ```
  - ```
    #navigation * {
      color: blue;
    }
    ```

The Power of Cascade

- When multiple styles or style sheets are used, they start to cascade and sometimes compete with one another due to CSS's inheritance feature
- Any tag on the page could potentially be affected by any of the tags surrounded by it

- So, which one wins? Nearest ancestor wins:
  - Inline style or directly applied style
  - The last style sheet declared in the `<head>` section

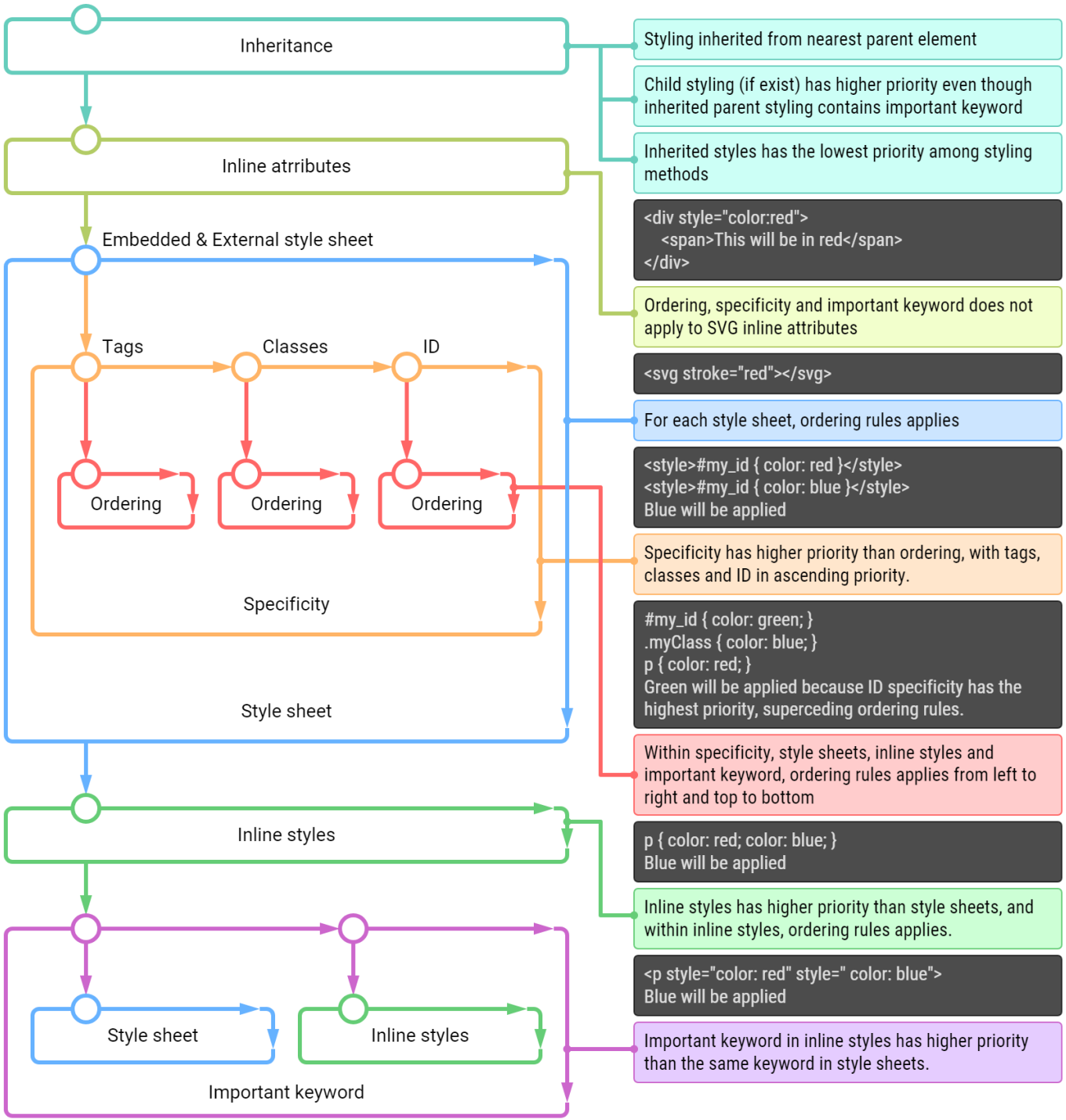
Inheritance

- Inheritance means CSS properties applied to one tag are passed on to nested tags
  - Only applied to properties whose value is `inherit`
  - ```
    <body style="font-family: Arial">
      This text will be Arial as well
    </body>
    ```

- Inherited property groups: font, spacing, text, list, visibility,…
- Non-inherited property groups: layout, box, background, alignment,…
    ```

Precedence

- By ascending order



Exercise

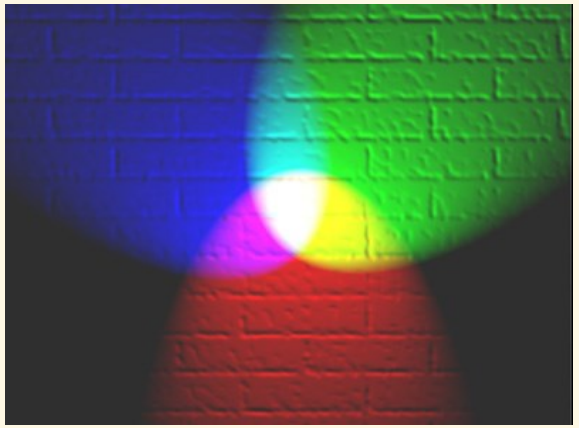
- Style tables as the followings. Pay attention to header row, odd/even rows, row on hover, “TYPE” column.

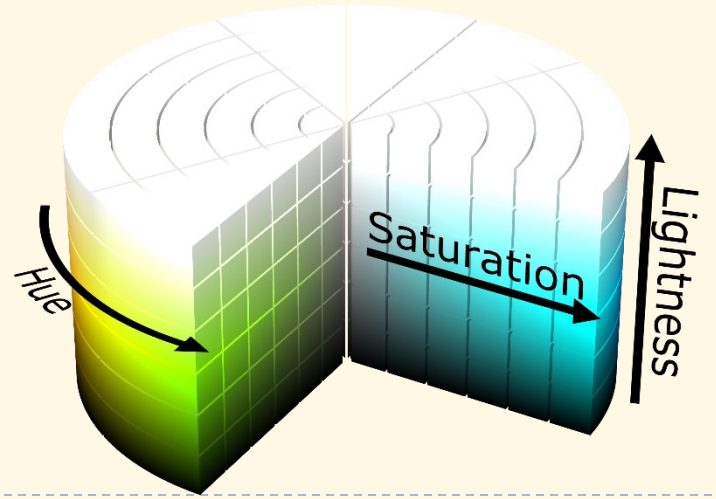
	TYPE	MESSAGE	USER
	cron	Cron run completed.	Anonymous (not verified)
	cron	Cron run completed.	Anonymous (not verified)
	system	views_ui module enabled.	admin
	system	views_ui module installed.	admin
	system	toolbar module enabled.	admin
	system	toolbar module installed.	admin
	system	breakpoint module	admin

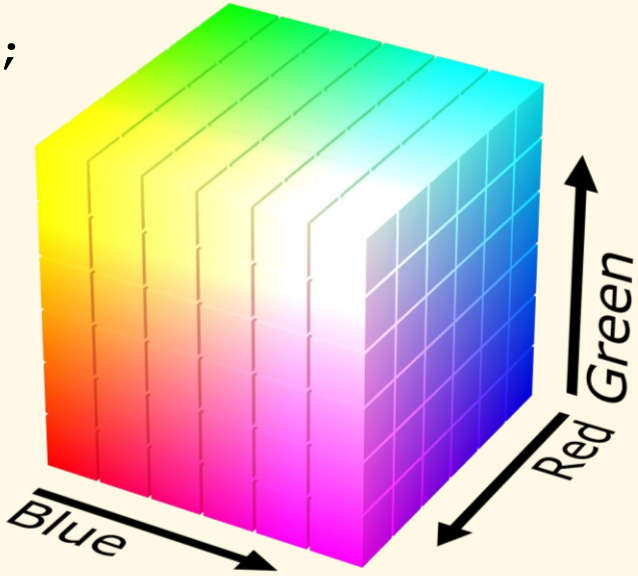
Text Formatting

Text Color

- Give color values to color property
  - `color: red;`
  - `color: #0f0;`
  - `color: #00ff00;`
  - `color: #00ff0080;`
  - `color: rgb(0, 127, 255);`
  - `color: rgba(0, 127, 255, 0.5);`
  - `color: hsl(120, 100%, 80%);`
  - `color: hsla(120, 100%, 80%, 0.5);`







Text Properties

- `text-align: left|right|center|justify;`
- `vertical-align: top|bottom|middle|baseline;`
- `text-decoration: underline|overline|line-through;`
- `text-transform: uppercase|lowercase|capitalize;`

- `text-indent: 20px;`
- `letter-spacing: 5px;`
- `word-spacing: 15px;`
- `line-height: 1.2;`

Font Properties

- `font: 15px Arial, sans-serif;`
- `font: italic small-caps bold 12px/1.2 Times;`

- `font-style: normal|italic;`
- `font-variant: normal|small-caps;`
- `font-weight: normal|bold|bolder|lighter|200|400|700;`
- `font-size: xx-small|x-small|smaller|small|medium|large|15px, 1em, 1rem;`
- `line-height: 1.2;`
- `font-family: "Times New Roman", Times, serif;`

CSS Length Units

- Absolute units:
  - `mm`, `cm`, `in`: millimeters, centimeters, inches
  - `pt`: points (1pt = 1/72 of 1in)
  - `px`: pixels
  - `pc`: picas (1pc = 12pt)
- Relative units:
  - `em`: relative to the font-size of the element (2em means 2 times the size of the current font)
  - `rem`: relative to the font-size of the root element (`<html>` element)
  - `%`: relative to the parent element
  - `vw`, `vh`: percentage relative to the browser window

Background Styling

Background Color

- Give color values to `background-color` property
  - `background-color: red;`
  - `background-color: #00ff00;`
  - `background-color: #00ff0080;`
  - `background-color: rgb(0, 127, 255);`
  - `background-color: rgba(0, 127, 255, 0.5);`
  - `background-color: hsl(120, 100%, 80%);`
  - `background-color: hsla(120, 100%, 80%, 0.5);`

Background Image

- `background: url(leaves.jpg) no-repeat top left;`

- `background-image: url(leaves.jpg);`
- `background-repeat: no-repeat|repeat|repeat-x|repeat-y;`
- `background-position: left|right|center|20%|10px top|center|bottom|30%|20px;`
- `background-size: auto|cover|contain|50%;`

- `background-origin: padding-box|border-box|content-box;`
- `background-clip: border-box|padding-box|content-box;`
- `background-attachment: scroll|fixed|local;`

Multiple Background Images

- Possible to put multiple images together in one background
- ```
  .multi-bg {
    background-image: url(decoration.png),
          url(ribbon.png), url(old_paper.jpg);
    background-repeat: no-repeat;
    background-position: left top,
          right bottom, left top;
  }
  ```


  ```

Inline Image

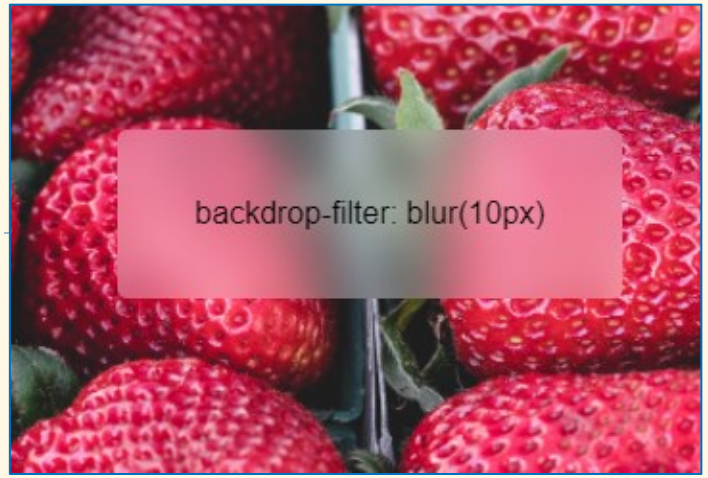
- Images can be embedded directly into CSS with Base64 encoding:
  - `background-image: url( AORHHOVSKudfOulrSOp3WOyDZu6QdvCchPGolfO0o/ XBs/fNwfjZ0frl3/zy7///wAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AACH5BAkAABAALAAAAAAQABAAAAVVICSOZGlCQAos J6mu7fiyZeKqNKToQGDsM8hBADgUXoGAiqhSvp5QAn QKGIgUhwFUYLCVDFCrKUE1lBavAViFIDlTImbKC5Gm 2hB0SlBCBMQiB0UjIQA7);`

Image in Text Background

- Currently CSS has no properties for image in text background
- Use the following trick:
  - ```
    .text {
      color: transparent;
      background-image: url(leaves.png);
      -webkit-background-clip: text;
    }
    ```

  - Attn: `background-clip` property is standard, but `text` value is not, hence use `-webkit-` prefix for WebKit-based browsers (Chrome, Edge, Safari,…)
    ```

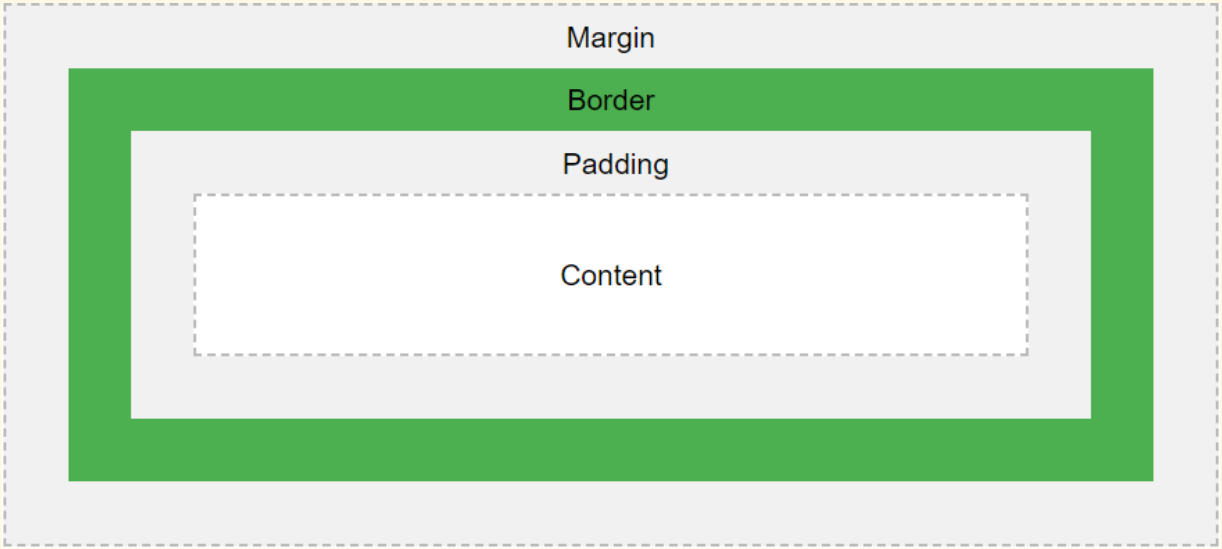
Backdrop as Background



- With `backdrop-filter` property:
  - `backdrop-filter: blur(2px);`
  - `backdrop-filter: contrast(40%);`
  - `backdrop-filter: grayscale(30%);`
  - `backdrop-filter: brightness(60%);`
  - `backdrop-filter: drop-shadow(4px 4px 10px blue);`
  - `backdrop-filter: hue-rotate(120deg);`
  - `backdrop-filter: invert(70%);`
  - `backdrop-filter: opacity(20%);`
  - `backdrop-filter: sepia(90%);`
  - `backdrop-filter: saturate(80%);`
- Multiple filters possible:
  - `backdrop-filter: blur(4px) saturate(150%);`

Box Model

Introduction



- All HTML elements can be considered as boxes
- The box model is a box consisting of:
  - Margin: Clears an area outside the border, and is transparent.
  - Border: Goes around the padding and content.
  - Padding: Clears an area around the content, and is transparent.
  - Content: Where text and child elements appear.

Box Sizing Modes

- `box-sizing: content-box|border-box;`

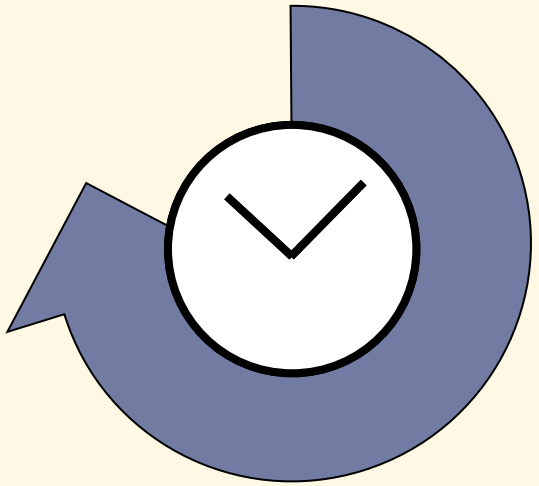
- `content-box`:
  - Default mode
  - Width and height properties include only the content
  - Actual size (w/h) = content + paddings + borders + margins
    - width, height = content size
- `border-box`:
  - Width and height properties include content, padding and border
  - Actual size (w/h) = width/height + margins
    - width, height = content + paddings + borders

Width and Height Properties

- Size of elements are controlled by:
  - `width: 200px|20em|30rem|40%|`
    `auto|fit-content;`
  - `height: 100px;`
  - `max-width,min-width: 500px|80%;`
  - `max-height,min-height: 500px;`
- `calc()` function:
  - `width: calc(10px + 100px);`
  - `max-width: calc(100% - 30px);`
  - `height: calc(2em * 5);`
  - Make sure that operators (+, −, *, /) are surrounded by whitespaces

Paddings

- `padding: auto;`
- `padding: 5px;`
- `padding: 10px 2rem;`
- `padding: 10px 0 1.2em 1rem;`



- `padding-top: 2px;`
- `padding-left: 2rem;`

Margins

- `margin: auto;`
- `margin: 5px;`
- `margin: 10px 2rem;`
- `margin: 10px 0 1.2em 1rem;`

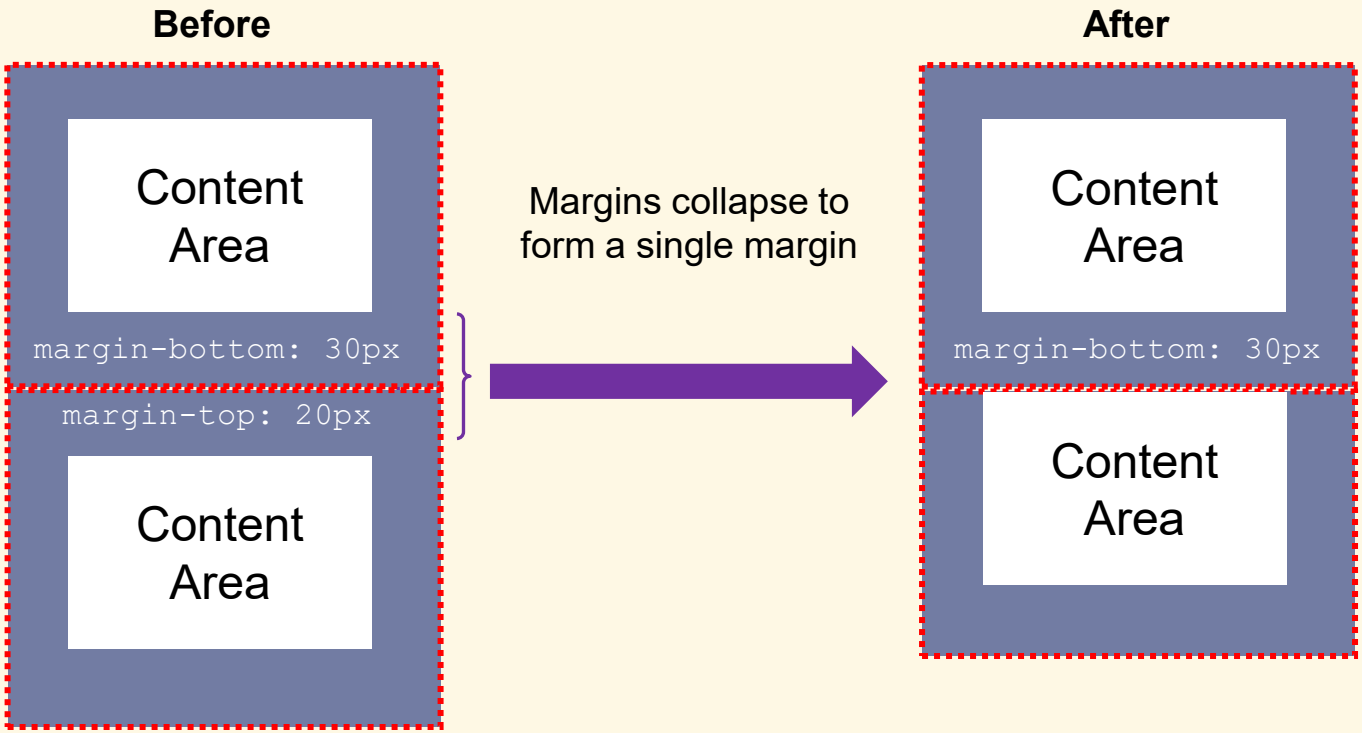
- `margin-top: 2px;`
- `margin-left: 2rem;`

Margin Collapse

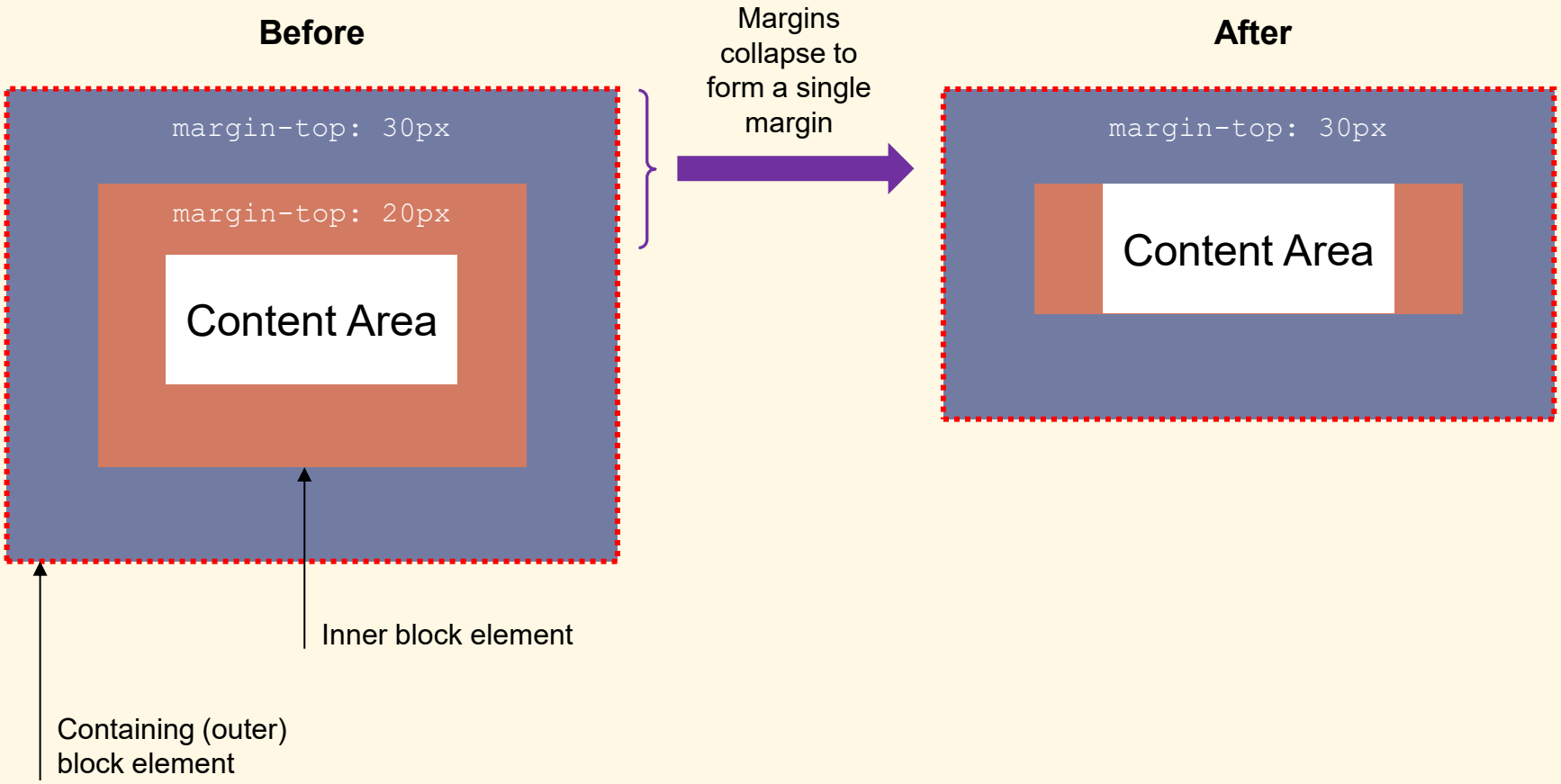
- When 2 or more vertical margins meet, they will collapse to form a single margin
  - The height of this combined margin is the largest of them
  - Only happens if there are no borders or padding separating the margins

- Margin collapse applies in 2 cases
  - 2 or more block elements are stacked one above the other
  - One block element is contained within another block element

Margin Collapse for Stacked Elements

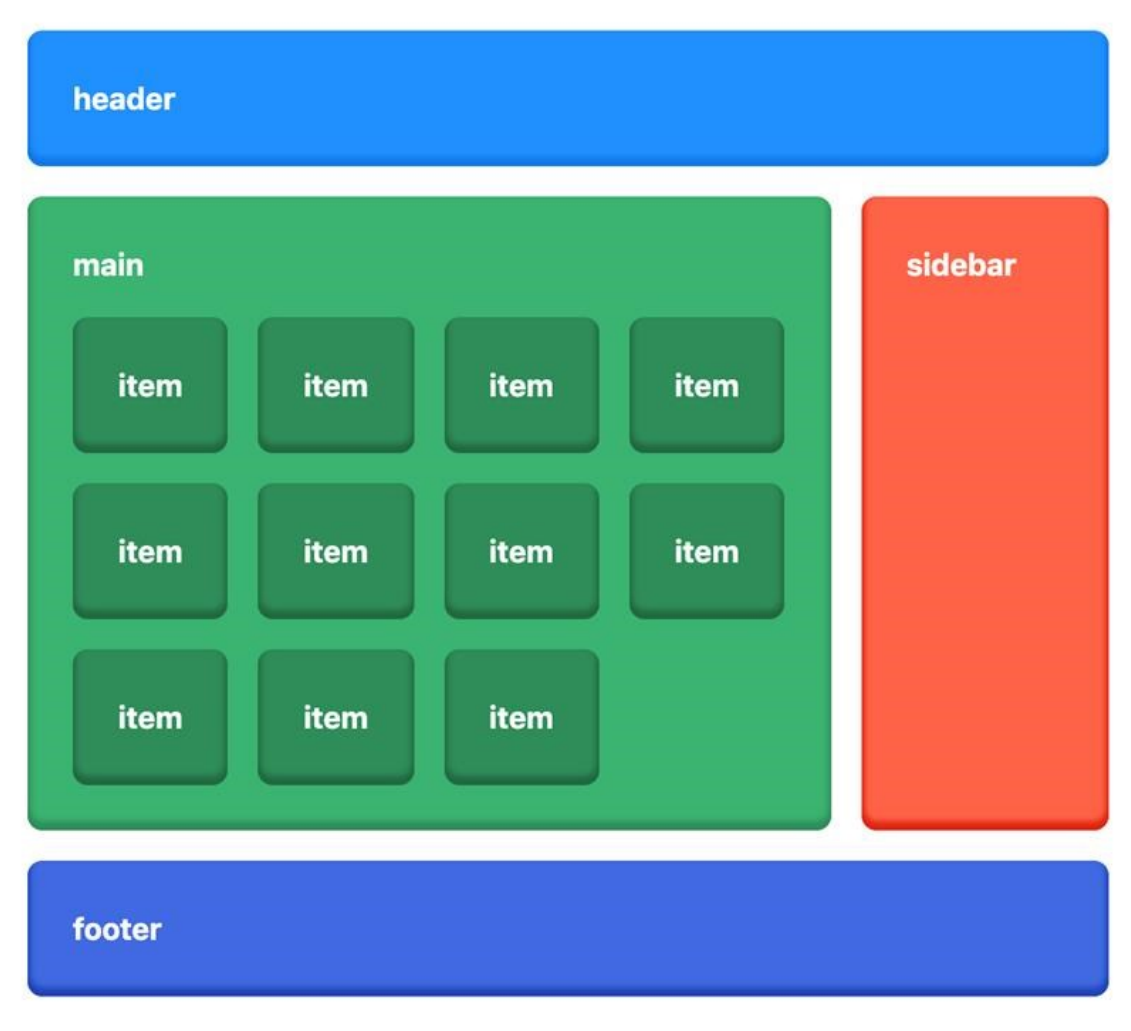


Margin Collapse for Contained Elements



Exercise

- Create a layout as in the figure using only `<div>` elements



Border Styling

Border Color

- `border-color: red;`
- `border-color: red #f29 rgb(0, 127, 255) blue;`
- `border-color: red blue;`

- `border-top-color: #00ff0080;`
- `border-right-color: rgba(0, 127, 255, 0.5);`
- `border-bottom-color: rgb(0, 127, 255);`
- `border-left-color: #00ff00;`

Border Style

- `border-style: solid|dotted|dashed|double|groove|inset|outset;`
- `border-style: solid dotted groove double;`
- `border-style: solid dashed;`
- `border-left-style: solid;`

- `border-width: 2px|medium|thick|thin;`
- `border-width: 1px 0 5px thin;`
- `border-width: 2px thin;`
- `border-right-width: 3px;`

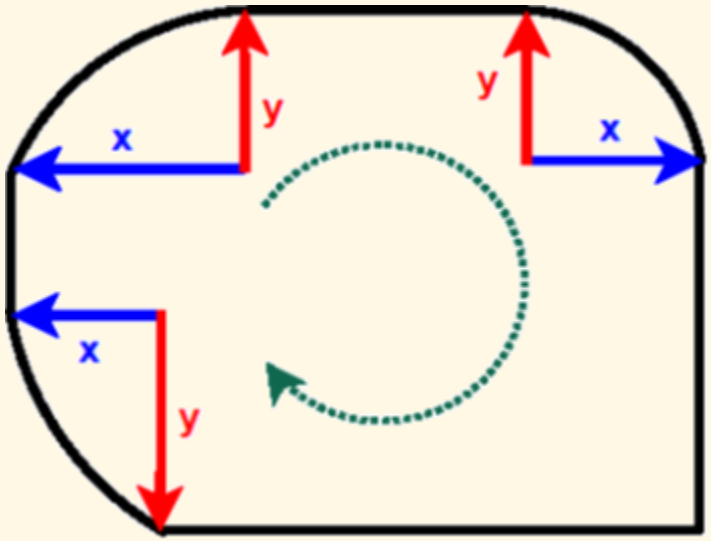
Border Shorthands

- `border: 5px solid red;`
- `border-top: 2px dotted #25f;`

Rounded Corners

- `border-radius: 5px|30%;`
- `border-radius: 2px 1rem 10% 3em;`
- `border-radius: 2px 3rem 0 4em / 1rem 2px 0 10%;`

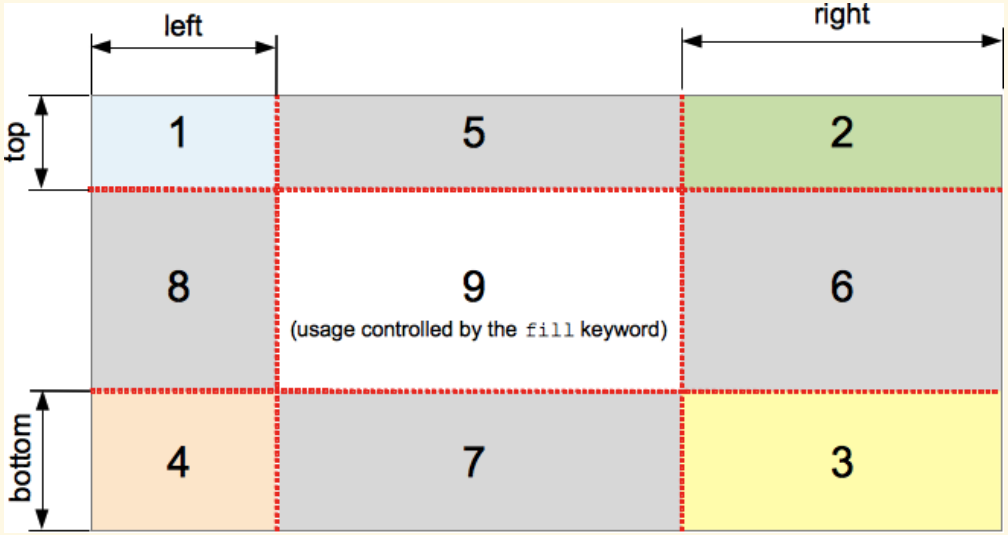
- `border-top-left-radius: 10px;`
- `border-top-right-radius: 10px 30%;`



Border Image

- `border-image: url(border.png) 30 round;`

- `border-image-source: url(border.png);`
- `border-image-slice: 30 20% fill 10;`



- `border-image-width: 10px;`
- `border-image-repeat: repeat|round|stretch|space;`

Outline

- An outline is a line that is drawn just outside the border edge of the elements
  - Generally used to indicate focus or active states of the elements such as buttons, links, form fields
  - Outlines don't take up space, may overlap other elements
  - Unlike borders, an outline is the same on all sides
  - Outlines may be non-rectangular, but you cannot create circular outlines
- `outline: 2px dotted green;`
- `outline-width: 1px|thick|thin|medium;`
- `outline-style: solid|dotted|dashed|ridge;`
- `outline-color: gray;`
- `outline-offset: 10px;`

Positioning & Display

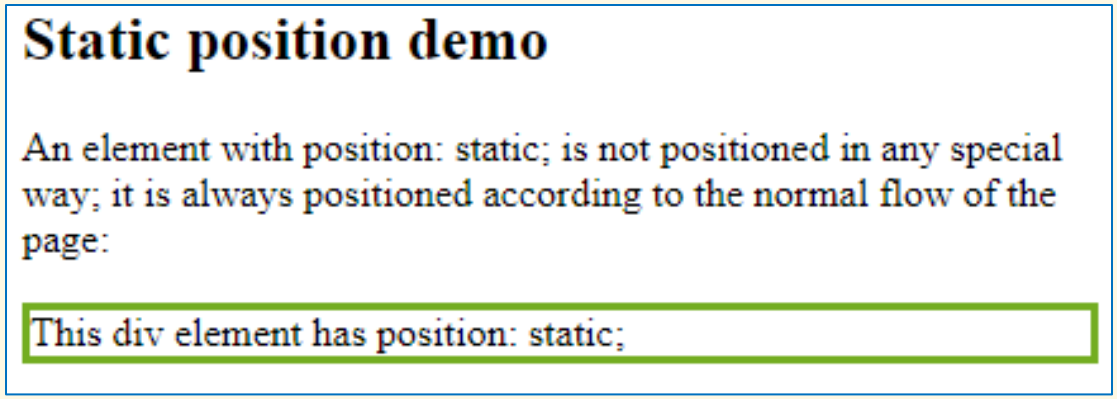
Positioning Methods

- `position` property:
  - `static` (default): Elements render in order, as they appear in the document flow
  - `absolute`: Element is positioned relative to its first positioned (not `static`) ancestor element
  - `fixed`: Element is positioned relative to the browser window
  - `relative`: Element is positioned relative to its normal position
  - `sticky`: Element is positioned based on the user's scroll position
- `position:absolute|relative|fixed|sticky` (except `static`) elements are called positioned

position: static

```
div.info {
  position: static;
  border: 3px solid #73AD21;
}
```

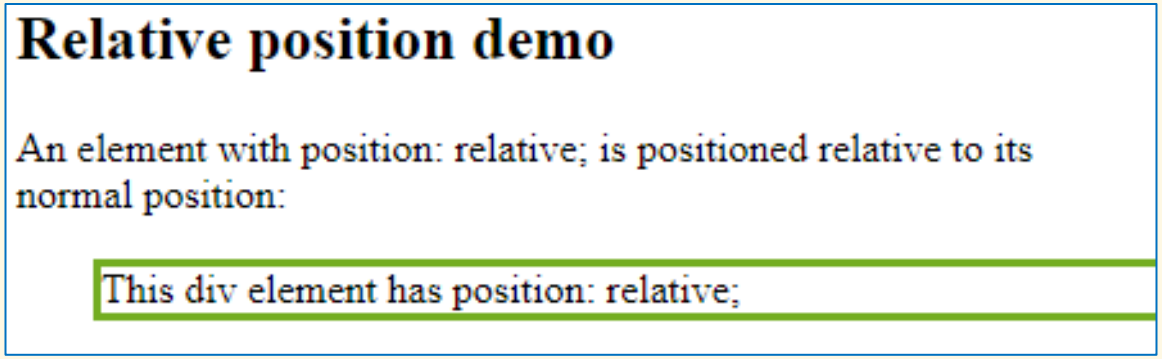
```
<div class="info">
  This div element has position: static;
</div>
```



position: relative

```
div.info {
  position: relative;
  left: 30px;
  border: 3px solid #73AD21;
}
```

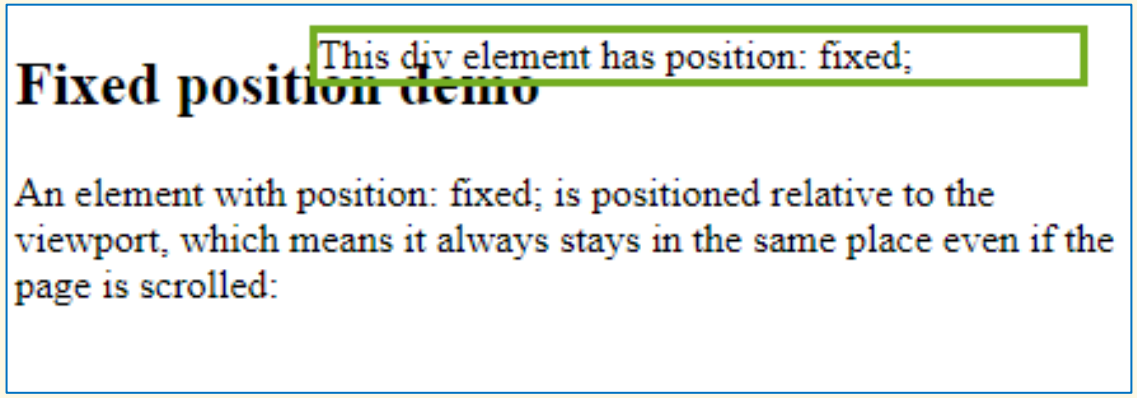
```
<div class="info">
  This div element has position: relative;
</div>
```



position: fixed

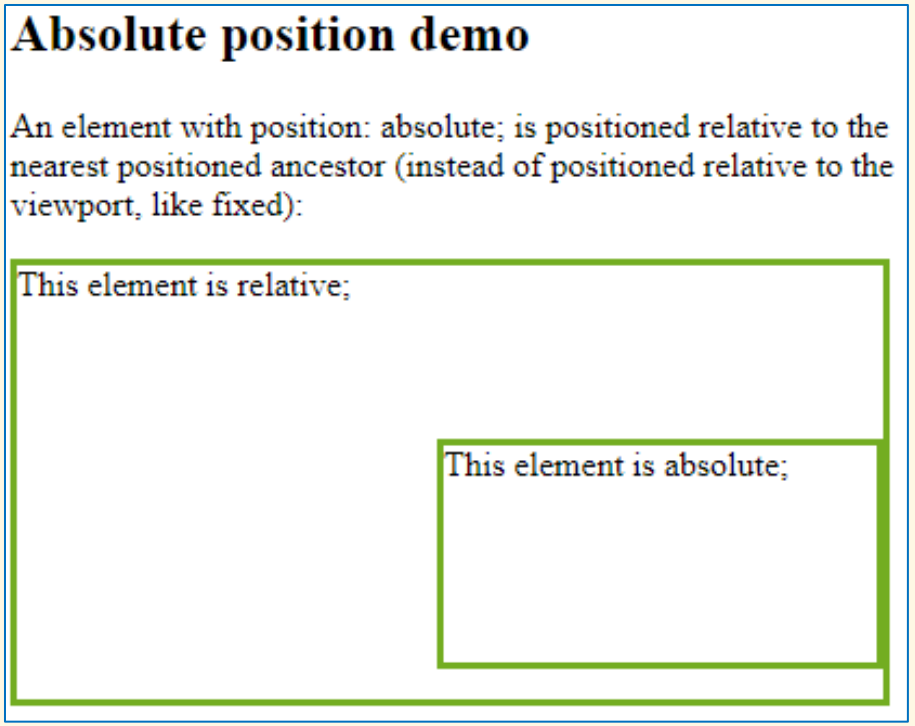
```
div.info {
  position: fixed;
  top: 10px;
  right: 20px;
  width: 300px;
  border: 3px solid #73AD21;
}
```

```
<div class="info">
  This div element has position: fixed;
</div>
```



position: absolute

```
div.parent {
  position: relative;
  width: 400px;
  height: 200px;
  border: 3px solid #73AD21;
}
div.child {
  position: absolute;
  top: 80px;
  right: 0;
  width: 200px;
  height: 100px;
  border: 3px solid #73AD21;
}
```

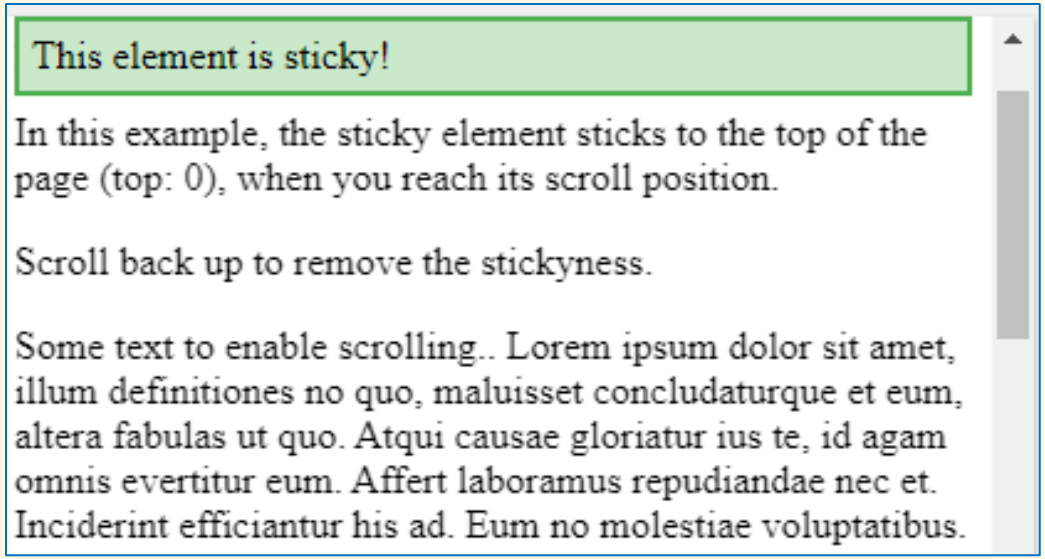



```
<div class="parent">This element is relative;
  <div class="child">This element is absolute;</div>
</div>
```

position: sticky

```
div.info {
  position: sticky;
  top: 0;
  padding: 5px;
  background-color: #cae8ca;
  border: 2px solid #4CAF50;
}
```

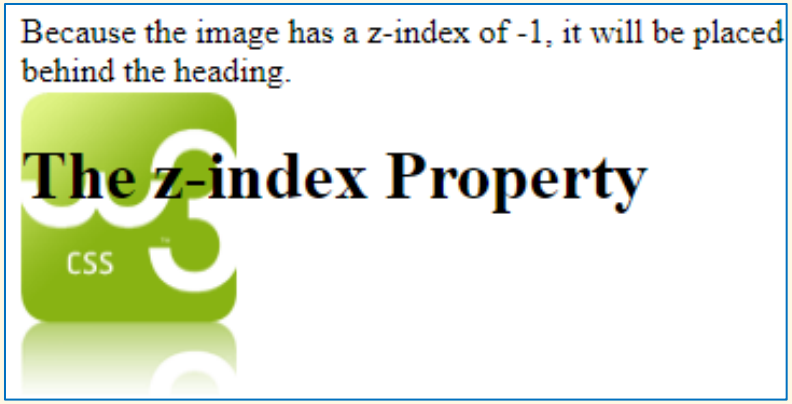
```
<div class="info">
  This element is sticky!
</div>
```



z-Index

- `z-index` property specifies the stack order of an element
  - Element with greater stack order is always in front of an element with a lower stack order
  - Only works on positioned elements

```
img {
  position: absolute;
  left: 0px;
  top: -20px;
  z-index: -1;
}
```




```
<div style="position: relative">
  <h1>The z-index Property</h1>
  <img src="w3css.gif" width="100" height="140">
</div>
```

Display Types

- `display` property:
  - `inline`: Element is inline (like `<span>`), any width and height properties have no effect
  - `block`: Element starts a new line and takes up the whole width (like `<div>`)
  - `inline-block`: Element is inline, but can take width and height properties
  - `none`: Element is not displayed
  - `list-item`: Element behaves like a `<li>`
  - `grid`: Element is a block-level grid container (discussed later)
  - `flex`: Element is a block-level flex container (discussed later)
  - `table`, `table-cell`, `table-row`, `table-column`,…

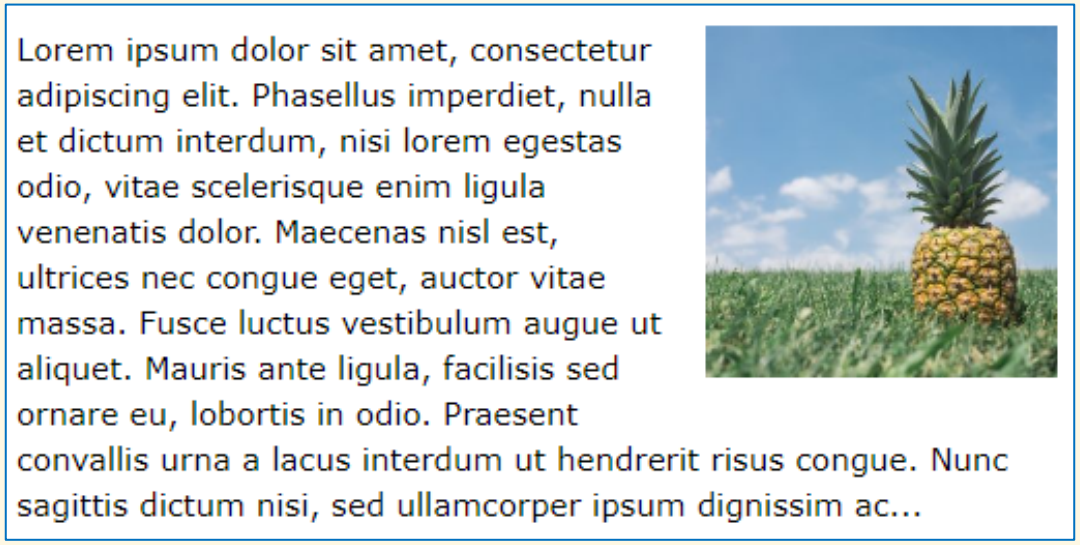
Visibility Methods

- `visibility` property:
  - `visible`: Element is displayed
  - `hidden`: Element is hidden

- "`visibility: hidden`" vs "`display: none`":
  - `visibility: hidden`
    - Takes up space in the layout
    - Children are still visible if they have "`visibility: visible`"
  - `display: none`
    - Does not take up space
    - Removes the element completely, including all children

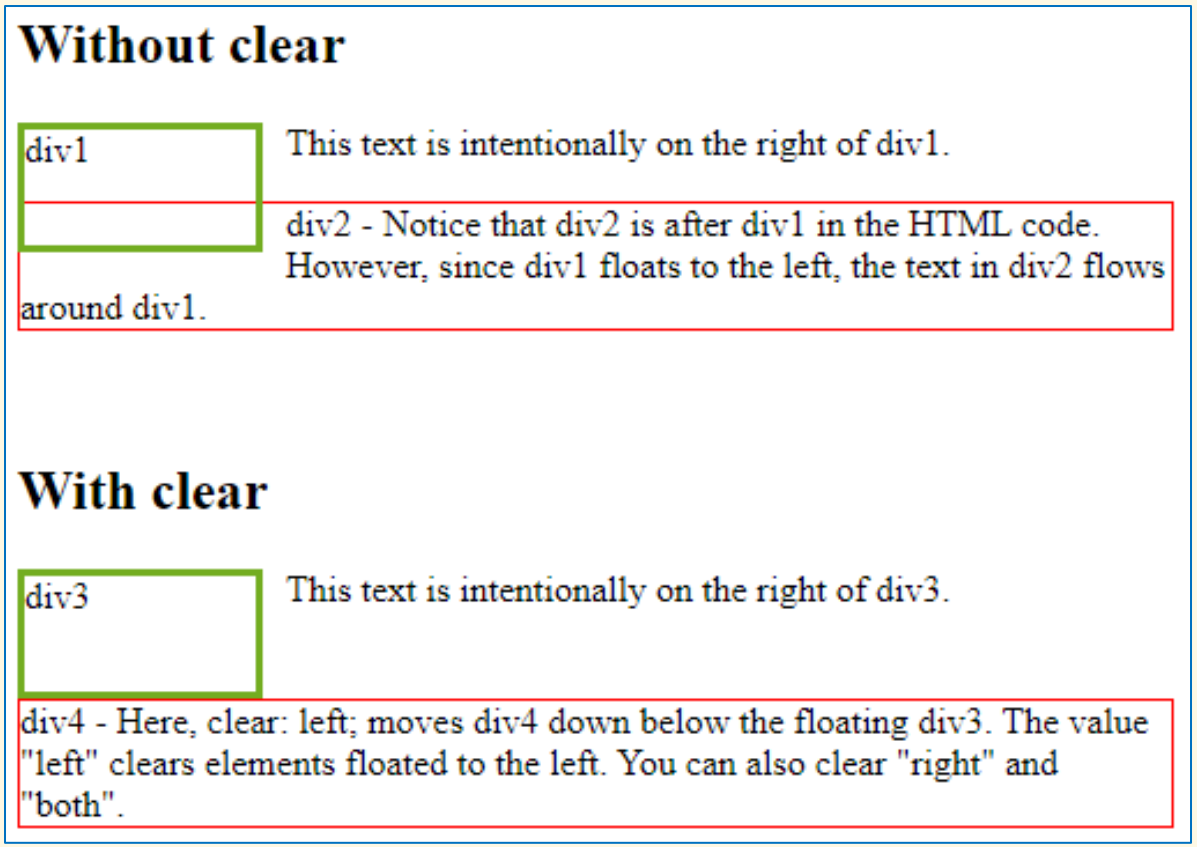
Float Methods

- Make elements float to one side of its container
  - `float: left|right;`
- Example:
  - ```
    img {
       float: right;
    }
    ```


    ```

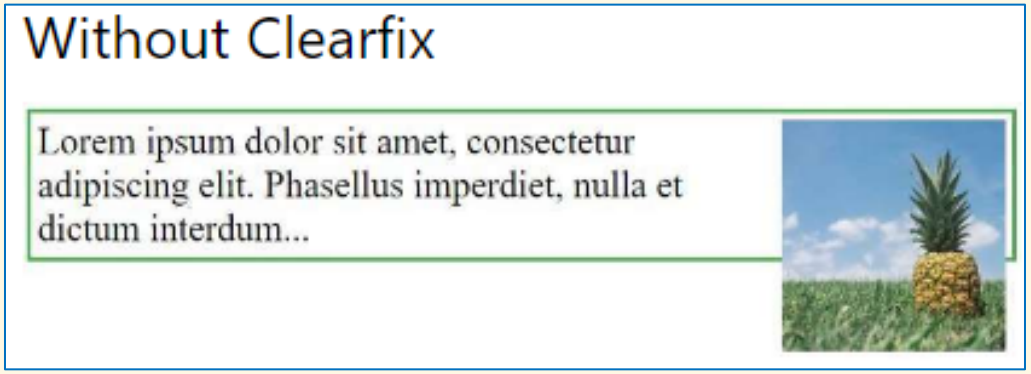
Clearing Floats

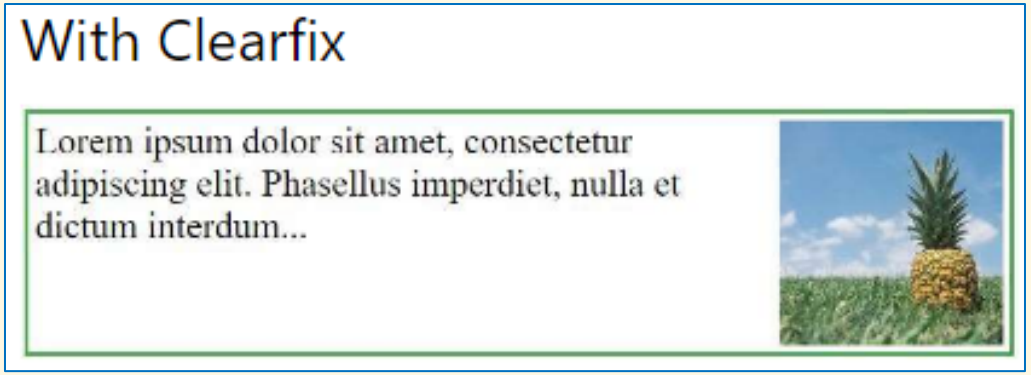
- Avoid floating elements on one or both sides
  - `clear: left|right|both;`



Clearfix

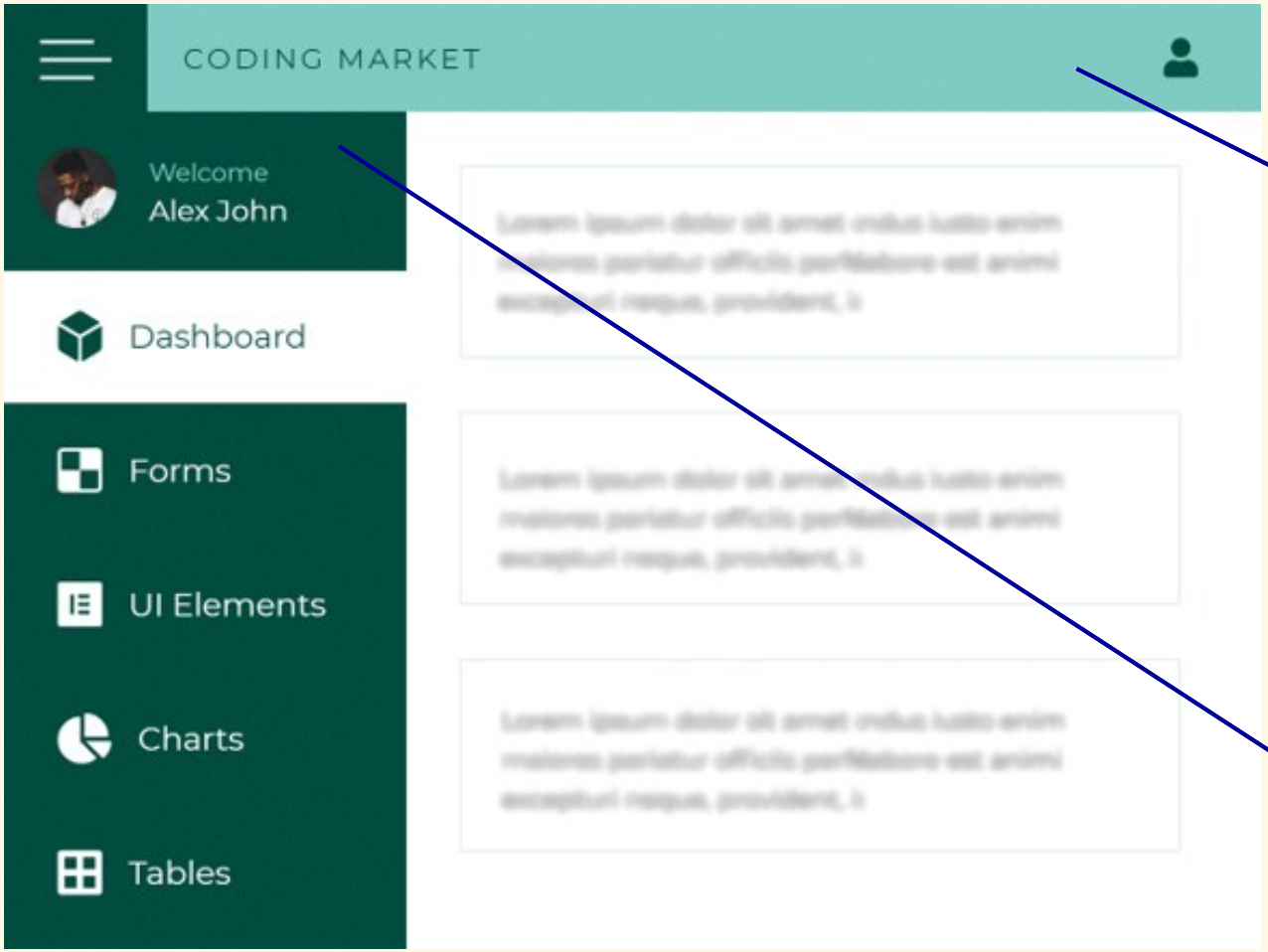
- Problem: If an element is taller than the element containing it, and it is floated, it will "overflow" outside of its container
- Solution: add "`overflow:auto`" to the containing element





Exercise

- Create a page with the following layout



Fixed header

Panel normally hidden, and shown when mouse pointer comes over the “hamburger button”

Pseudo-classes

Introduction

- Pseudo-classes select regular elements but under certain conditions

- Types of pseudo-classes:
  - Dynamic pseudo-classes
  - State pseudo-classes
  - Range pseudo-classes
  - Structural pseudo-classes
  - Matching pseudo-classes
  - Other pseudo-classes

Dynamic Pseudo-classes

- Unvisited link:
  - `a:link`
- Visited link:
  - `a:visited`
- Selected link:
  - `a:active`
- Mouse-over:
  - `:hover`
- Focused element:
  - `:focus`
  - `:focus-within`

State Pseudo-classes

- Form control states:
  - `:disabled`
  - `:enabled`
- `<input>` states:
  - `:checked`
  - `:read-only`
  - `:read-write`
  - `:required`
  - `:optional`
  - `:valid`
  - `:invalid`

Range Pseudo-classes

- First child:
  - `:first-child`
- Last child:
  - `:last-child`
- n^{th} child:
  - `:nth-child(n [of selector])`
  - `p:nth-child(5)`
  - `p:nth-child(2 of .highlighted)`
  - `p:nth-child(odd)`
  - `p:nth-child(even)`
  - `p:nth-child(2n+1)`
- n^{th} last child:
  - `:nth-last-child(n)`
- n^{th} child of type:
  - `:nth-of-type(n)`
- n^{th} last child of type:
  - `:nth-last-of-type(n)`

Structural Pseudo-classes

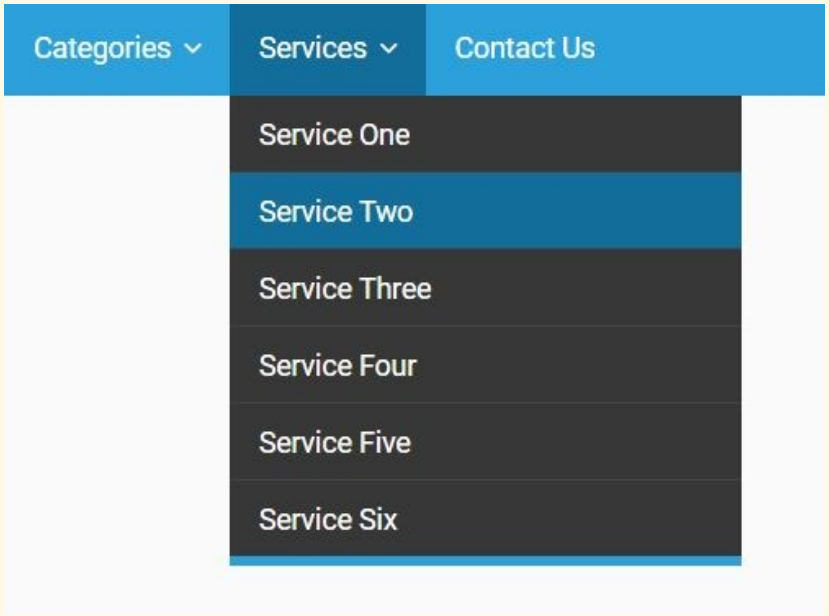
- Only child:
  - `:only-child`
- Only of type:
  - `:only-of-type`
- Root:
  - `:root`
- Empty:
  - `:empty`
- Target element:
  - `:target`

Matching Pseudo-classes

- Negation: elements not matching the given selector
  - `:not(.intro) { }`
  - `.intro:not(:first-child) { }`
  - `p:not(.intro):not([title^="flower"]) { }`
- Matches-any: elements matching one of the given selectors
  - `:is(header, .intro, :first-child) a { }`
  - `.button:is(a, button) { }`
  - `p:is(.header, .footer):not(.dark) { }`
- Containment: elements containing specific children
  - `h1:has(p) { }`
  - `h1:has(h2):has(> p) { }`

Exercises

- Create a button group that looks like the figure
- Create a menubar with dropdown submenus



Pseudo-elements

Introduction

- Pseudo-elements effectively create new elements that are not specified in the markup of the document and can be manipulated much like a regular element
- Pseudo-elements:
  - `::before`
  - `::after`
  - `::first-letter`
  - `::first-line`

::before and ::after

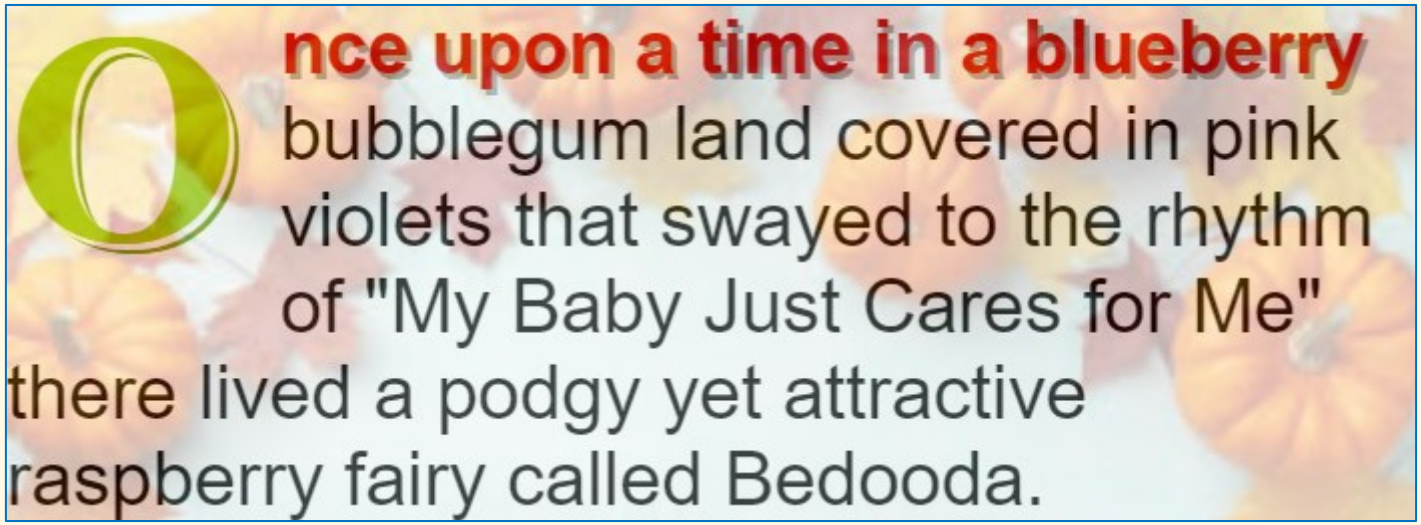
- Insert some content before and after the content of the specified elements
  - ```
    .intro::before {
      content: "Read this:";
      display: block;
      color: green;
    }
    .intro::after {
      content: "End of introduction";
      display: block;
      color: yellow;
    }
    ```
    ```

::first-letter and ::first-line

- Used to add a style to the first letter or the first line of the specified selector
  - ```
    p::first-letter {
      font-size: 200%;
      color: #8A2BE2;
    }
    ```
  - ```
    p::first-line {
      color: #ff0000;
      font-variant: small-caps;
    }
    ```

Exercise

- Style a paragraph as in the following:



Effects

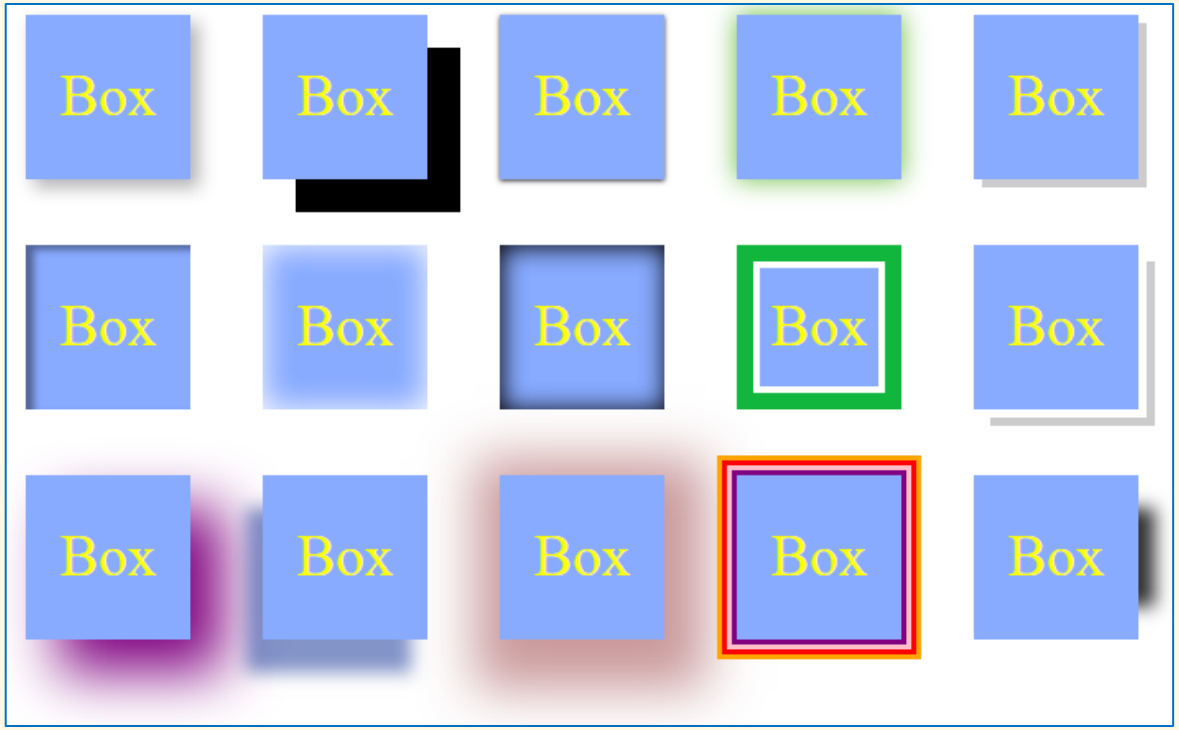
Text Shadow

- Syntax: `text-shadow: x y b c;`
  - *x* = horizontal offset
    - *x* < 0: left of the text
    - *x* > 0: right of the text
  - *y* = vertical offset
    - *y* < 0: above the text
    - *y* > 0: below the text
  - *b* = blur radius
  - *c* = shadow color

- Examples:
  - `text-shadow: 2px 3px 5px gray;`
  - `text-shadow: 1px 1px 1px #000, 3px 3px 5px blue;`

Box Shadow

- Syntax: `box-shadow: [inset] x y b c;`
  - *x* = horizontal offset
    - *x* < 0: left of the box
    - *x* > 0: right of the box
  - *y* = vertical offset
    - *y* < 0: above the box
    - *y* > 0: below the box
  - *b* = blur radius
  - *c* = shadow color



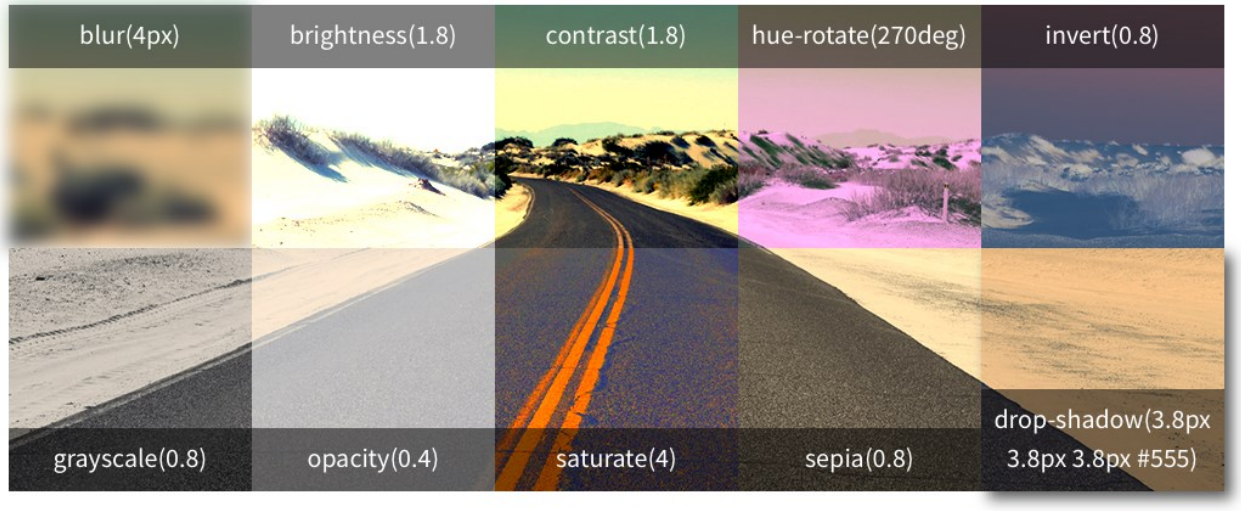
- Examples:
  - `box-shadow: 2px 3px 5px gray;`
  - `box-shadow: 1px 1px 1px #000, 3px 3px 5px blue;`

Opacity

- Opacity makes the affected elements transparent

Filters

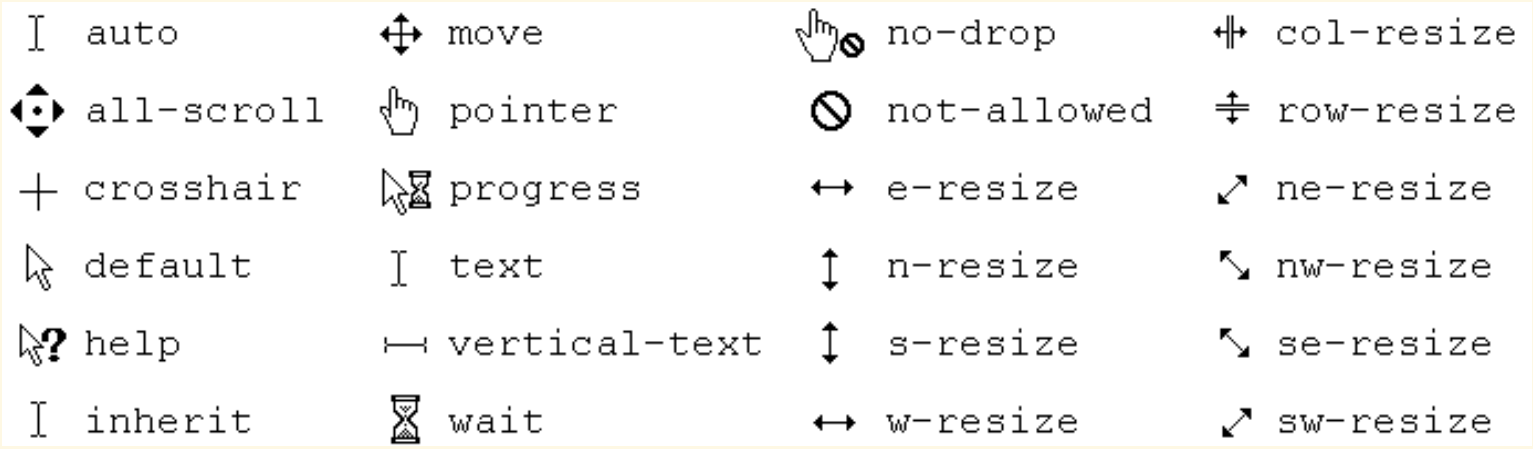
- More effects with `filter` property:
  - `opacity(0.5)`
  - `blur(5px)`
  - `contrast(200%)`
  - `grayscale(80%)`
  - `invert(70%)`
  - `saturate(30%)`
  - `sepia(60%)`
  - `brightness(0.4)`
  - `hue-rotate(90deg)`
  - `drop-shadow(2px 3px 5px red)`



- Examples:
  - `filter: blur(5px);`
  - `filter: contrast(175%) blur(3px) grayscale(30%);`

Cursor

- Specifies the cursor type to be shown when user moves the pointer over the element
  - `cursor: default | none | context-menu | help | pointer | progress | wait | cell | crosshair | text | vertical-text | alias | copy | move | no-drop | not-allowed | grab | grabbing | e-resize | n-resize | ne-resize | nw-resize | s-resize | se-resize | sw-resize | w-resize | ew-resize | ns-resize | nesw-resize | nwse-resize | col-resize | row-resize | all-scroll | zoom-in | zoom-out | url(`cursor-image-file`)`



Exercise

- Create a button with shadow and pressed effect on mouse click

Gradients

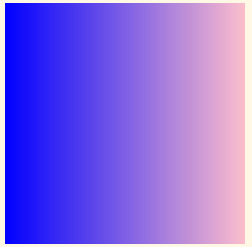
Gradients

- Gradients can be used in place of images, which can be:
  - `background-image`, `border-image-source`, `list-style-image`, `cursor`
- Types:
  - Linear
  - Radial
  - Conic
- Examples:
  - `background-image: linear-gradient(red, #2f8, orange);`
  - `border-image-source: radial-gradient(#f88, green);`
  - `list-style-image: conic-gradient(violet, blue);`

Linear Gradients: Direction

- `background: linear-gradient(blue, pink);`

- `background: linear-gradient(to right, blue, pink);`

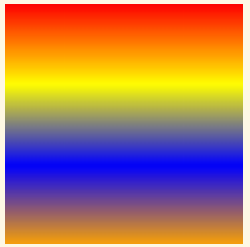


- `background: linear-gradient(to bottom right, blue, pink);`

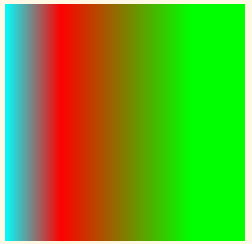
- `background: linear-gradient(70deg, blue, pink);`

Linear Gradients: Color Stops

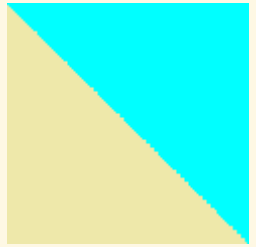
- `background: linear-gradient(red, yellow, blue, orange);`



- `background: linear-gradient(to left, lime 28px, red 77%, cyan);`

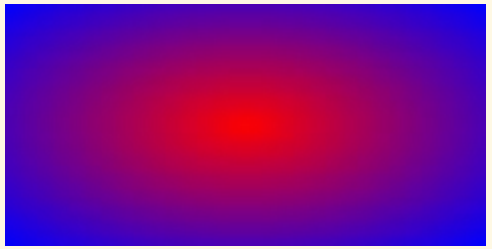


- `background: linear-gradient(to bottom left, cyan 50%, palegoldenrod 50%);`

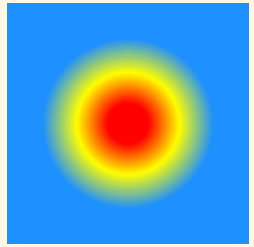


Radial Gradients

- `background: radial-gradient(red, blue);`



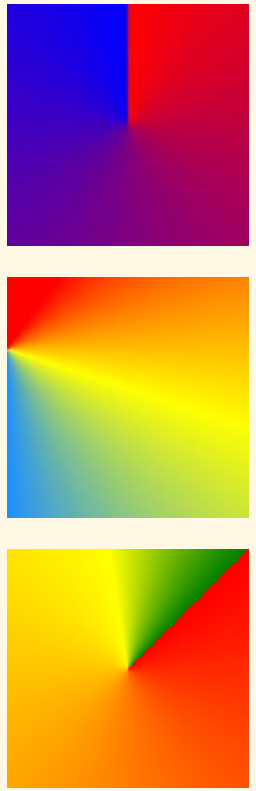
- `background: radial-gradient(red 10px, yellow 30%, #1e90ff 50%);`



- `background: radial-gradient(at 0% 30%, red 10px, yellow 30%, #1e90ff 50%);`

Conic Gradients

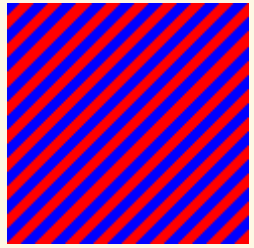
- `background: conic-gradient(red, blue);`
- `background: conic-gradient(at 0% 30%, red 10%, yellow 30%, #1e90ff 50%);`
- `background: conic-gradient(from 45deg, red, orange 50%, yellow 85%, green);`



Repeating Gradients

- Use `repeating-linear-gradient()`, `repeating-radial-gradient()`, and `repeating-conic-gradient()` instead

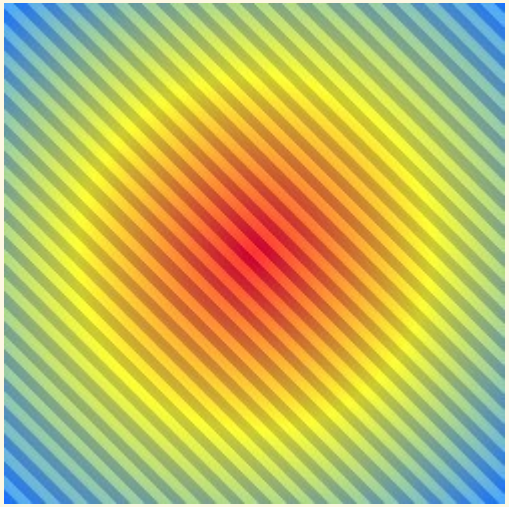
-
```
background: repeating-linear-gradient(-45deg, red, red 5px, blue 5px, blue 10px);
```



-
```
background: repeating-radial-gradient(#e66465, #9198e5 20%);
```

Multiple Gradients

```
background:
  radial-gradient(
    rgba(255, 0, 0, 0.8),
    rgba(255, 255, 0, 0.8),
    rgba(30, 144, 255, 0.8)),
  repeating-linear-gradient(45deg,
    rgba(0, 0, 255, 1),
    rgba(0, 0, 255, 1) 5px,
    rgba(255, 255, 255, 0) 5px,
    rgba(255, 255, 255, 0) 10px);
```



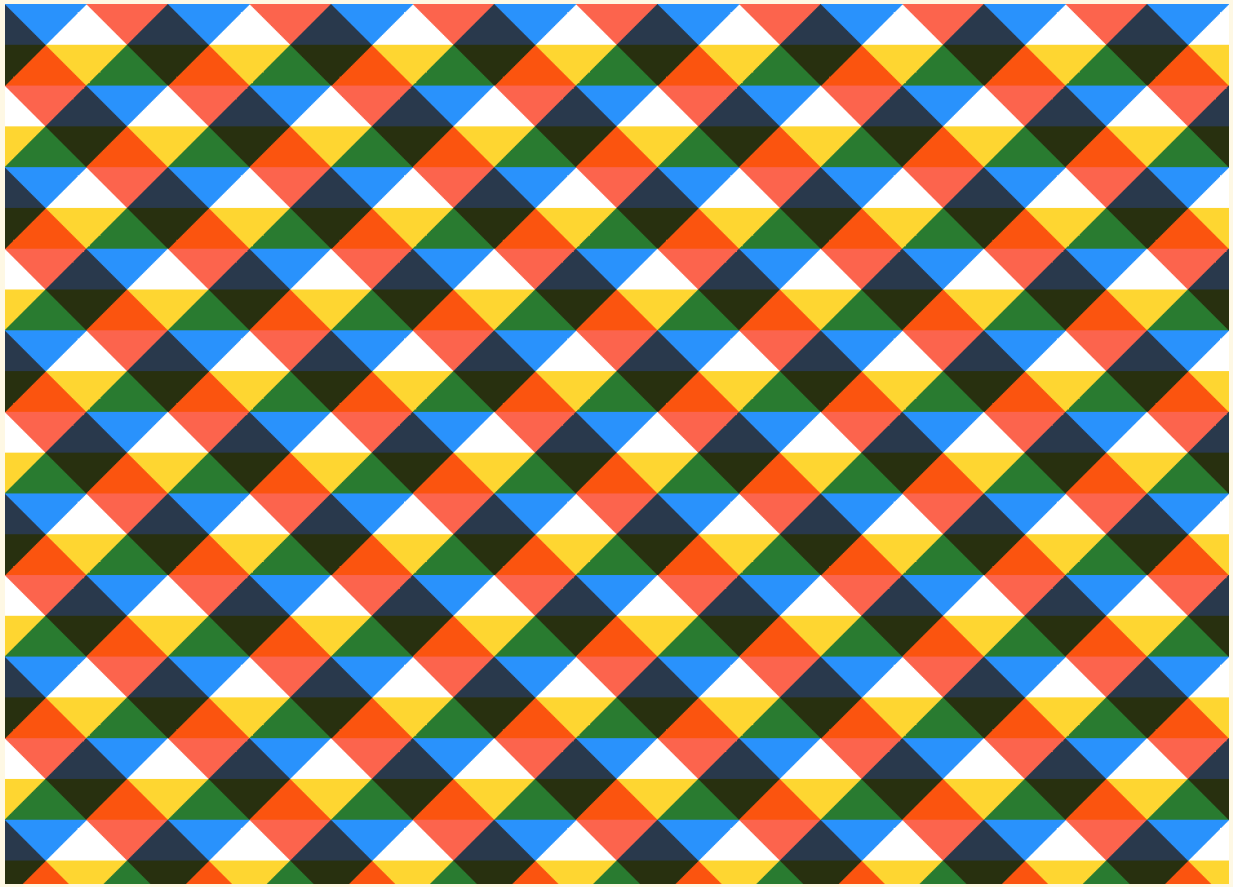
Text Gradients

- Currently CSS has no properties for text gradients
- Use the same trick as for image in text background:

```
.text {
    color: transparent;
    background-image: linear-gradient(
                to right, red, blue, green);
    -webkit-background-clip: text;
}
```

Exercise

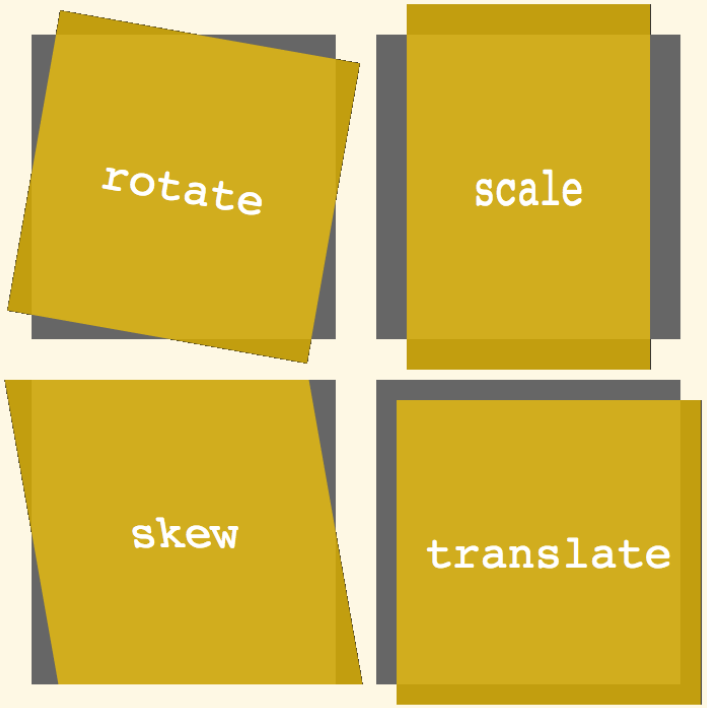
- Create a background with gradients as in the following



Transformations

2D Transformations

- Use `transform` property with following methods
  - `translate(dx, dy)`
  - `translateX(dx)`
  - `translateY(dy)`
  - `rotate(ang)`
  - `scale(sx, sy)`
  - `scaleX(sx)`
  - `scaleY(sy)`
  - `skew(angX, angY)`
  - `skewX(ang)`
  - `skewY(ang)`
  - `matrix(m11, m12, m21, m22, dx, dy)`

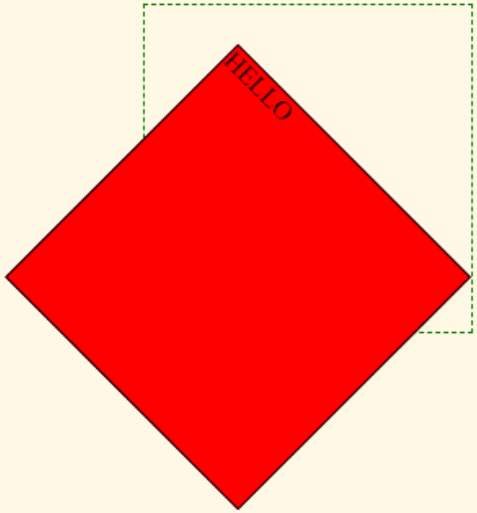


- `transform-origin`: changes the position on transformed elements

Examples

- `transform: rotate(45deg) scale(2);`

- `transform: rotate(45deg);`
  `transform-origin: 0 40%;`



3D Transformations

- Use `transform` property with following methods
  - `translate3d(`*dx*`, `*dy*`, `*dz*`)`
  - `rotate3d(`*x*`, `*y*`, `*z*`, `*a*`)`
  - `scale3d(`*sx*`, `*sy*`, `*sz*`)`
  - `perspective(`*d*`)`
  - `matrix3d(...)`

Transitions

Transitions

- Effects that let an element gradually change from one style to another
- Example

```
div {
  width: 200px;
  transition: width 2s;
}

div:hover {
  width: 300px;
}
```

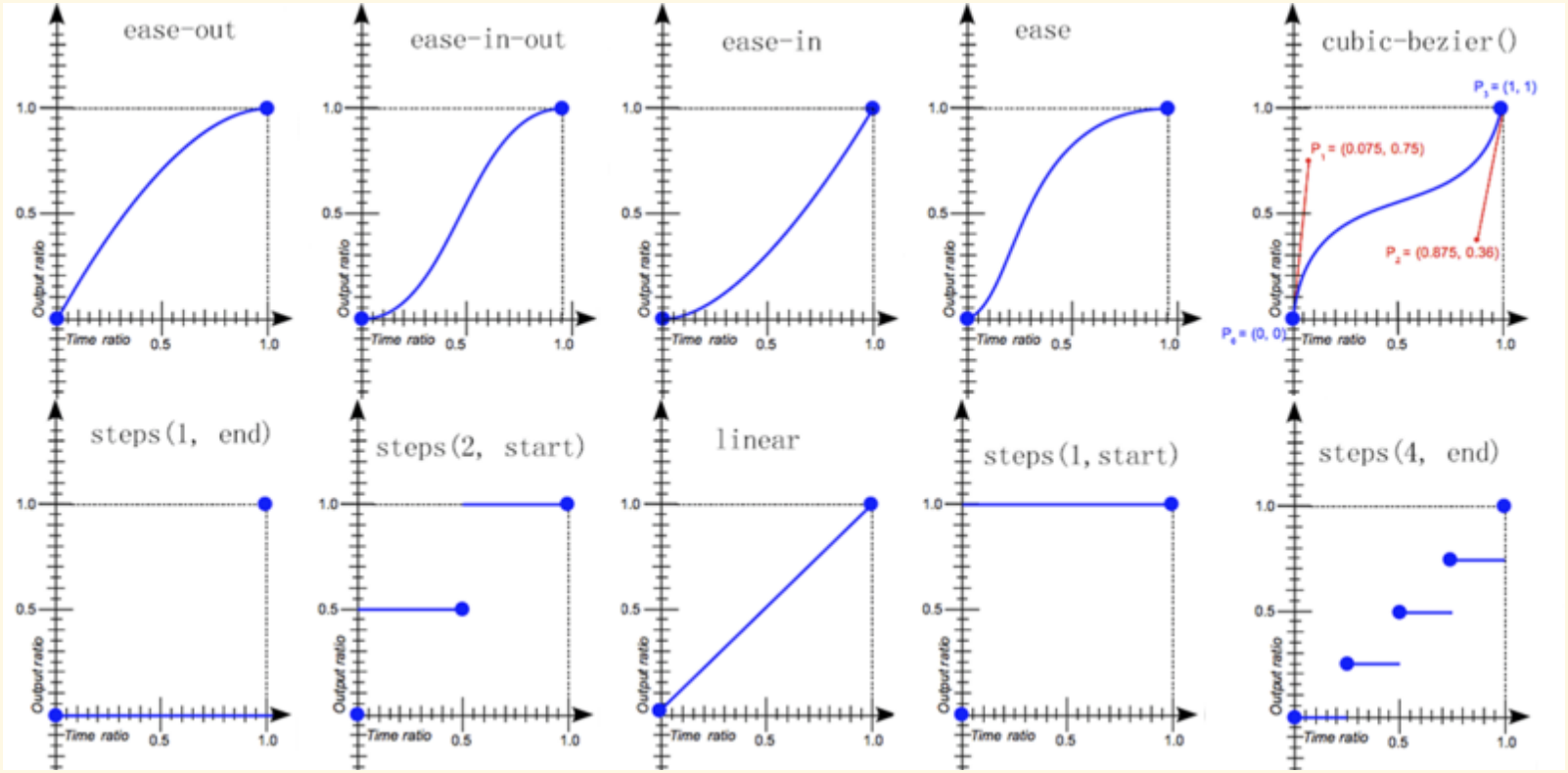
- Multiple transitions at once:
  - `transition: width 2s, height 1s, transform 1.5s;`

Detailed Properties and Shorthand

- Detailed:
  - ```
    transition-property: width;
    transition-duration: 1s;
    transition-timing-function: linear;
    transition-delay: 2s;
    ```
- Equivalent shorthand:
  - ```
    transition: width 1s linear 2s;
    ```

transition-timing-function

- linear|ease|ease-in|ease-out|ease-in-out
- steps(*steps*, start|end)
- cubic-bezier(*n*, *n*, *n*, *n*)



Animatable Properties

- Not all properties can be animated
- In general, the animatable properties are related to:
  - Color
  - Background
  - Border
  - Outline
  - Position
  - Margin, padding, size
  - Text
  - Opacity
  - Transforms, filters

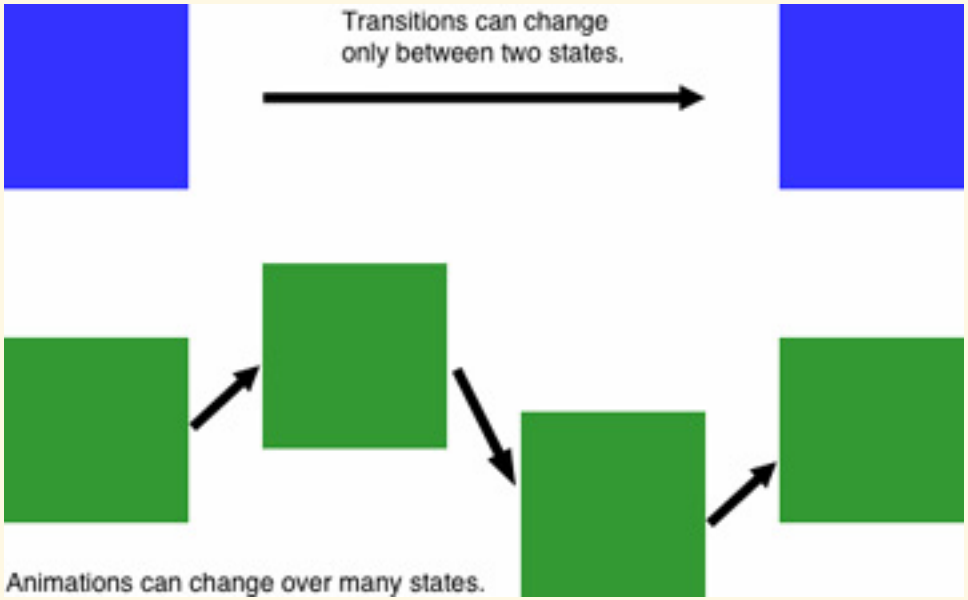
Exercises

- Create a button that changes its background color, text color, border with transitions on mouse events

Animations

Introduction

- Animations let an element gradually change from one style to another
  - Possible to change multiple CSS properties, multiple times
- Animations vs transitions
  - Transitions require a trigger (hover, focus,…) to run; animations don't
  - Transitions are limited to an initial and final state; animation can include many intermediate states (called keyframes)
  - Animations can loop, be started, stopped, paused dynamically



Keyframes

- Keyframes hold the styles the element has at certain times
- Example

```
@keyframes example {
  from {background-color: red;}
  to {background-color: yellow;}
}

div {
  animation-name: example;
  animation-duration: 4s;
}
```

Timing

```
@keyframes example {
    0%   {background-color: red;}
    25%  {background-color: yellow;}
    50%  {background-color: blue;}
    100% {background-color: green;}
}
```

- Attn:
  - `from` = 0%
  - `to` = 100%

Play Options

- Delayed animation:
  - `animation-delay: 2s;`
  - If using negative value, the animation will start as if it had already been playing a while
- Repeated animation:
  - `animation-iteration-count: 3;`
  - `animation-iteration-count: infinite;`
- Animation play direction:
  - `animation-direction: normal;` (forwards - default)
  - `animation-direction: reverse;` (backwards)
  - `animation-direction: alternate;` (fwds then bwds)
  - `animation-direction: alternate-reverse;` (bwds then fwds)
- Timing functions
  - `animation-timing-function` is used similarly to `transition-timing-function`

Animation Shorthand

- Shorthand
  - `animation: example 5s linear 2s infinite alternate;`

- Equivalence
  - ```
    animation-name: example;
    animation-duration: 5s;
    animation-timing-function: linear;
    animation-delay: 2s;
    animation-iteration-count: infinite;
    animation-direction: alternate;
    ```
    ```

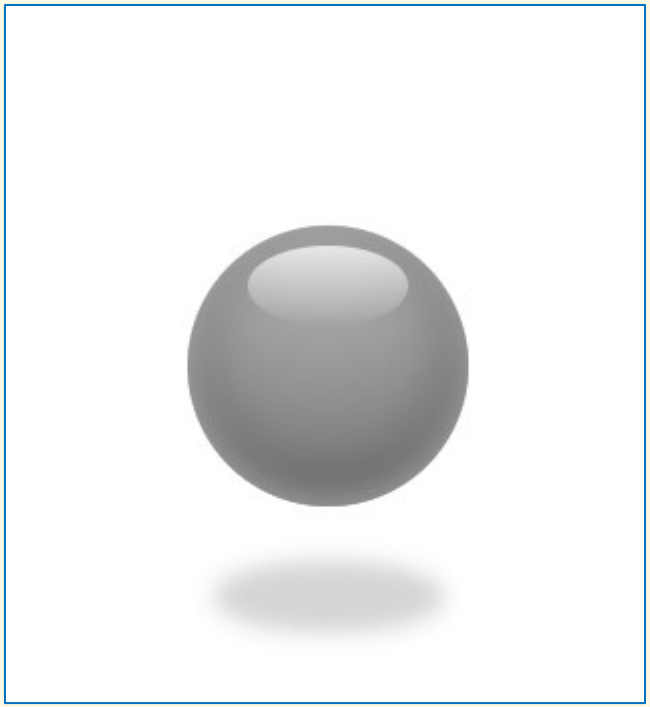
Pausing and Resuming an Animation

- Animation play state:
  - `animation-play-state: paused | running`
- Example:
  - ```
    div {
      animation-name: example;
      animation-duration: 4s;
      animation-direction: alternate;
      animation-iteration-count: infinite;
    }

    div:hover {
      animation-play-state: paused;
    }
    ```
    ```

Exercise

- Create and style a bouncing ball with animation as in the following:



Media Queries and Responsive Design

Motivation

- Document should look good on any device (desktop, tablet, phone) → responsive (dynamic) layouts
- Solution: Use CSS to shrink, enlarge, hide or move content in order to look good on any screen
- How to do:
  - Setting viewport:
    - `<meta name="viewport" content="width=device-width, initial-scale=1.0">`
  - Do not rely on a particular viewpoint
  - Use relative dimensions (`%`): `width`, `height`
  - Use relative units (`em`, `rem`)
  - Use flexbox for layout
  - Use media-queries to apply different styles to large/small screens

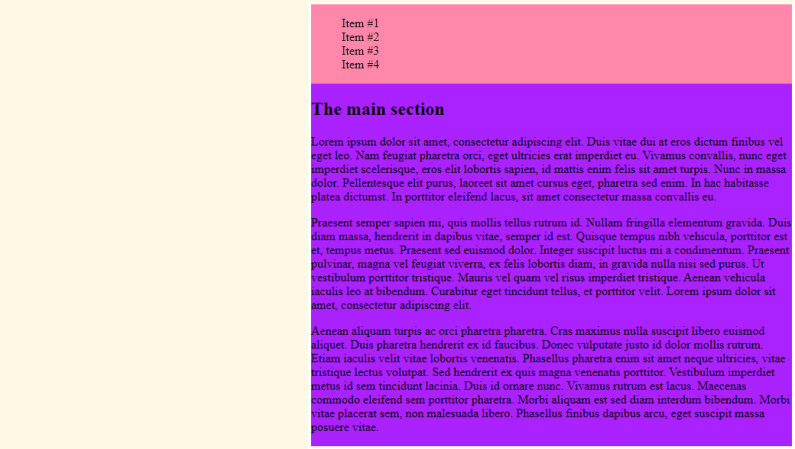
Media Queries

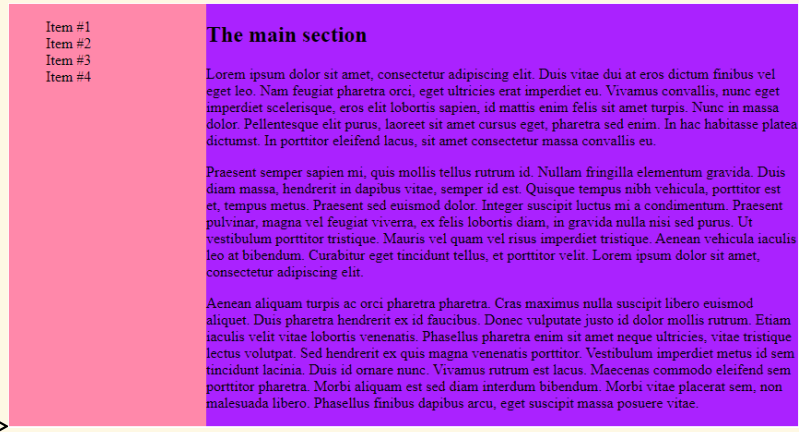
- Use `@media` to apply CSS style only if a condition is met
  - ```
    @media only screen and (max-width:500px) {
      #div1 {
        width: 100%;
      }
    }
    ```
  - `@media only screen and (min-width:500px) { }`
  - `@media only screen and (orientation:landscape) { }`
    ```

Media Query Example

```
body  { display: flex; }
#left { width: 25%; }
#main { width: 75%; }

@media screen and (max-width:
800px) {
  body {
    flex-direction: column;
  }
  #left, #main {
    width: 100%;
  }
}
```

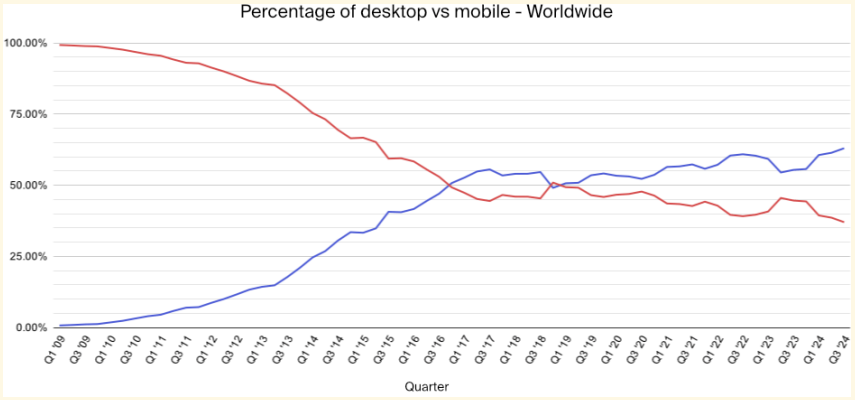





```
<section id="left">…</section>
<section id="main">…</section>
```

Mobile-First Design

- Mobile Internet use passed desktop in Oct 2016
- Mobile-first design = design for mobile before designing for desktop or any other device
  - This will make the page display faster on smaller devices
  - Ex: Instead of changing styles when the width gets smaller than 768px, we should change the design when the width gets larger than 768px



Typical Device Breakpoints

- Extra small devices: phones (≤ 600px)
  - `@media only screen and (max-width: 600px) { }`
- Small devices: portrait tablets and large phones (≥ 600px)
  - `@media only screen and (min-width: 600px) { }`
- Medium devices: landscape tablets (≥ 768px)
  - `@media only screen and (min-width: 768px) { }`
- Large devices: laptops, desktops (≥ 992px)
  - `@media only screen and (min-width: 992px) { }`
- Extra large devices: large laptops and desktops (≥ 1200px)
  - `@media only screen and (min-width: 1200px) { }`

General Syntax

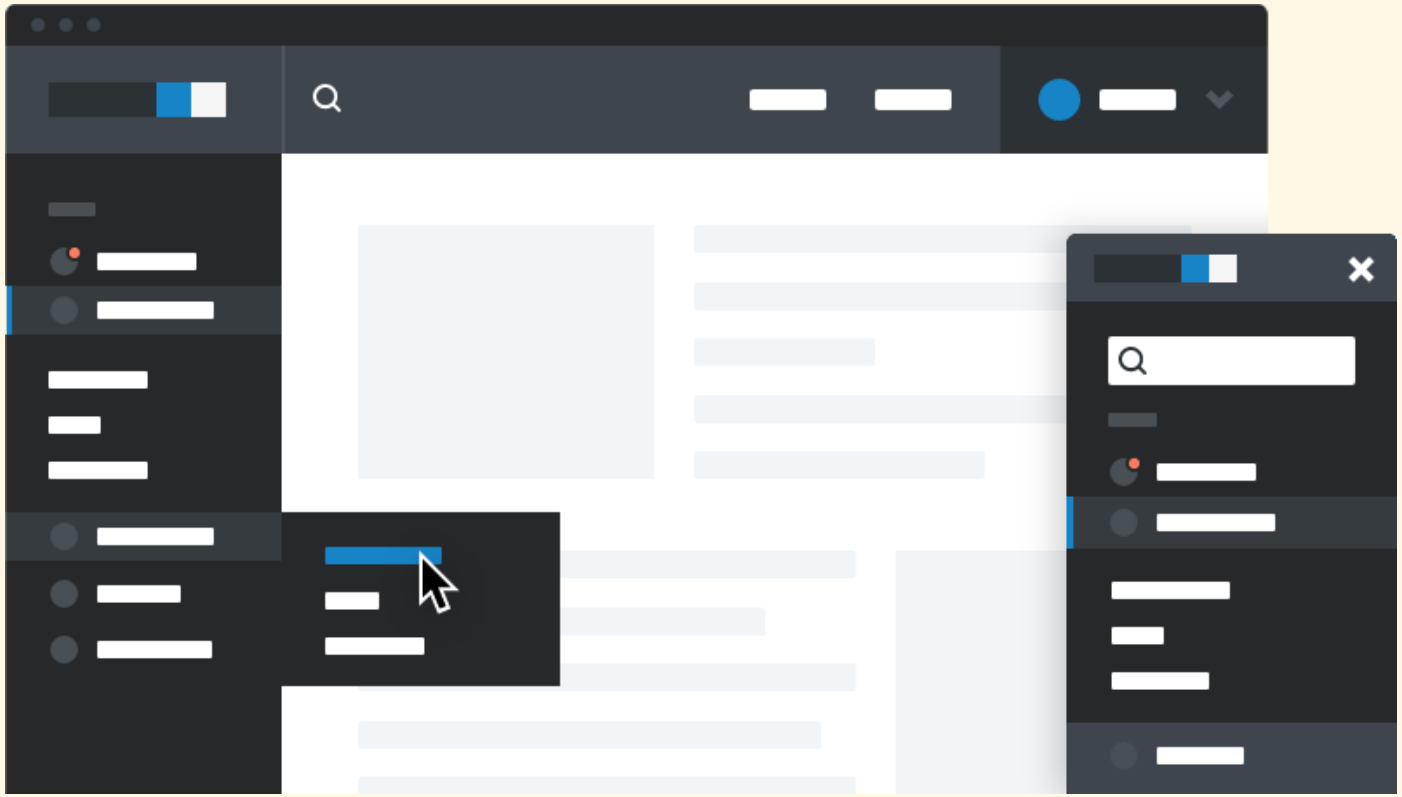
- Syntax:
  - `@media [not|only] mediatype and (mediafeature and|or|not mediafeature) { }`
  - `mediatype: all|screen|print|speech`
  - `mediafeature: width|max-width|min-width| orientation|...`
- Examples
  - `@media only screen and (orientation:landscape) { }`
  - `@media print { }`
  - `@media screen and (max-width: 900px) and (min-width: 600px), (min-width: 1100px) { }`
    (When the width is between 600px and 900px, or above 1100px)

Conditional CSS Attachment

- External CSS stylesheets can be applied with conditions
  - `<link rel="stylesheet" type="text/css" href="screen.css" media="screen" />`
  - `<link rel="stylesheet" type="text/css" href="print.css" media="print" />`
  - `<link rel="stylesheet" type="text/css" href="main_1.css" media="screen and (max-device-width: 480px)" />`

Exercise

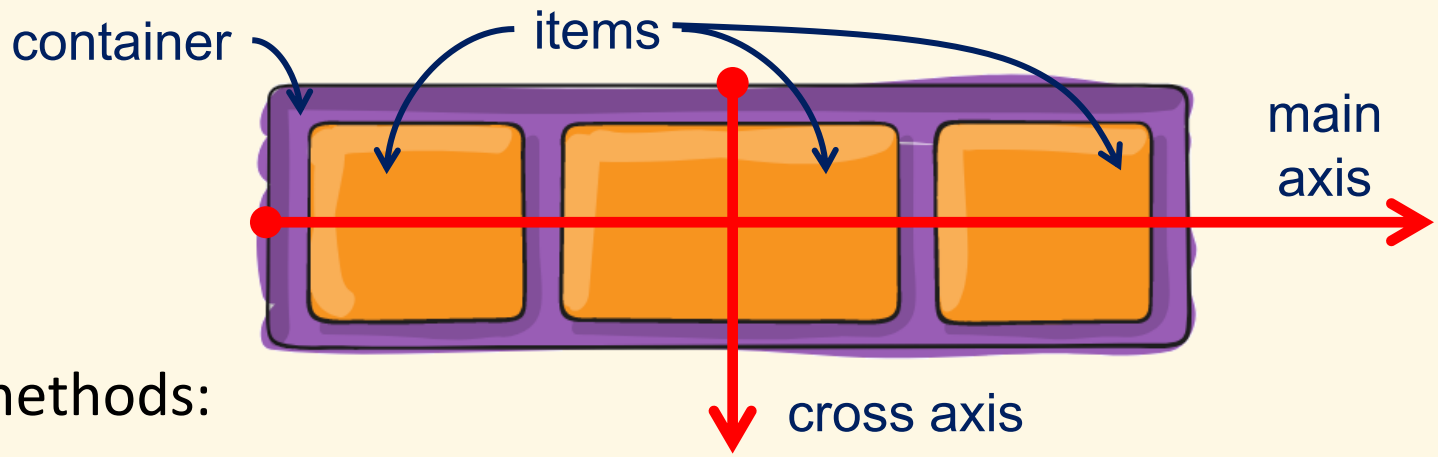
- Design the following responsive page layout with a menubar and a sidebar



Flexbox Layout

Introduction

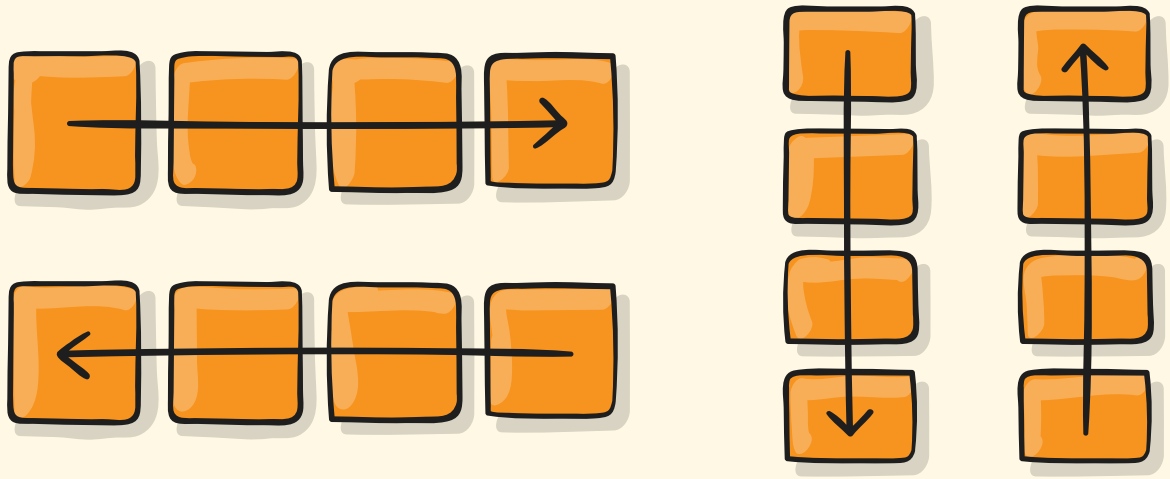
- Flexbox (flexible box) aims at providing a more efficient way to define layouts
- More flexibility to support complex page structures
- A flexbox consists of a container (parent) and its items (children)



- Display methods:
  - ```
    .container {
      display: flex; /* or inline-flex */
    }
    ```
    ```

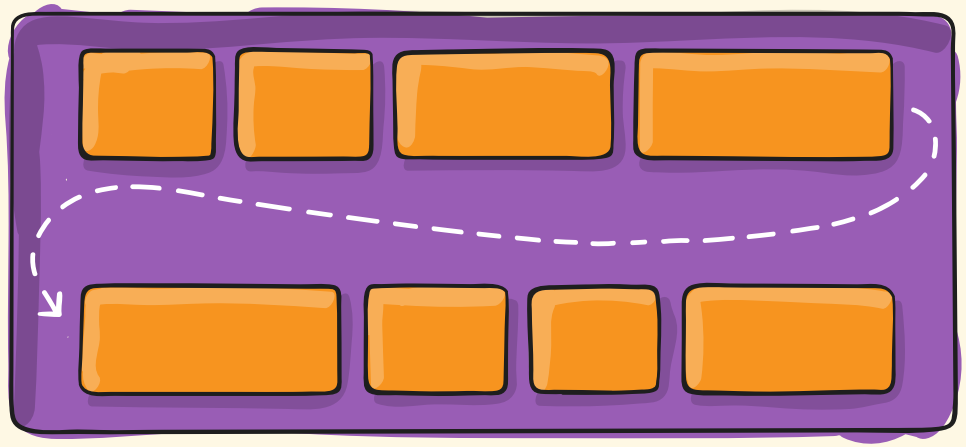
Container: `flex-direction`

- `row` (default): left to right in ltr; right to left in rtl
- `row-reverse`: right to left in ltr; left to right in rtl
- `column`: same as `row` but top to bottom
- `column-reverse`: same as `row-reverse` but bottom to top



Container: `flex-wrap`

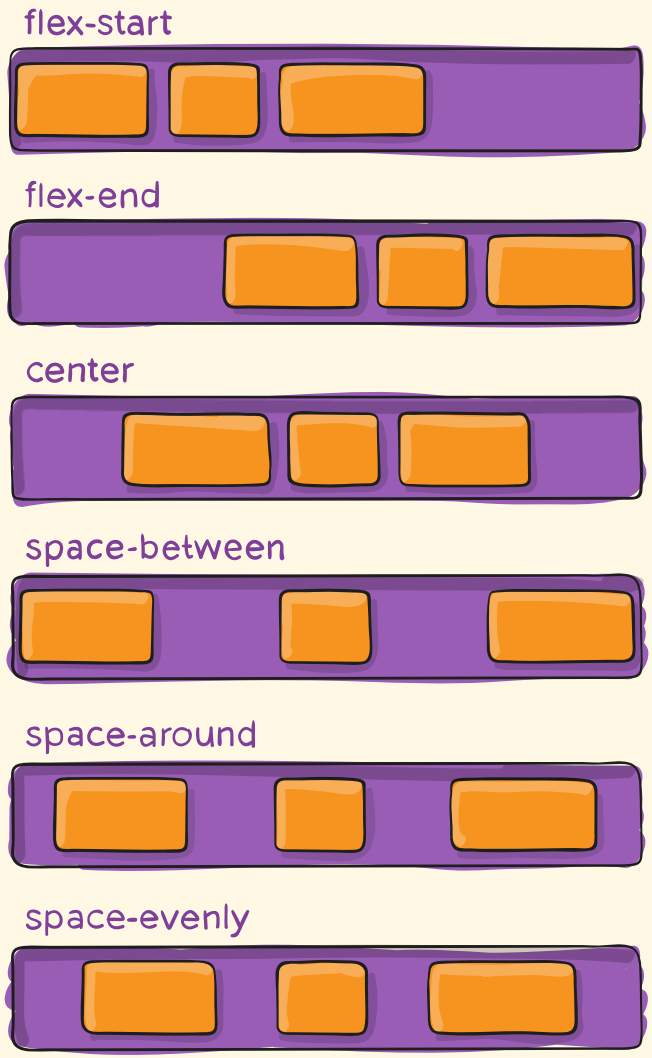
- `nowrap` (default): all items will be on one line
- `wrap`: items will wrap onto multiple lines, from top to bottom
- `wrap-reverse`: items will wrap onto multiple lines from bottom to top



- Shorthand for `flex-direction` and `flex-wrap`:
  - `flex-flow: column-reverse wrap;`

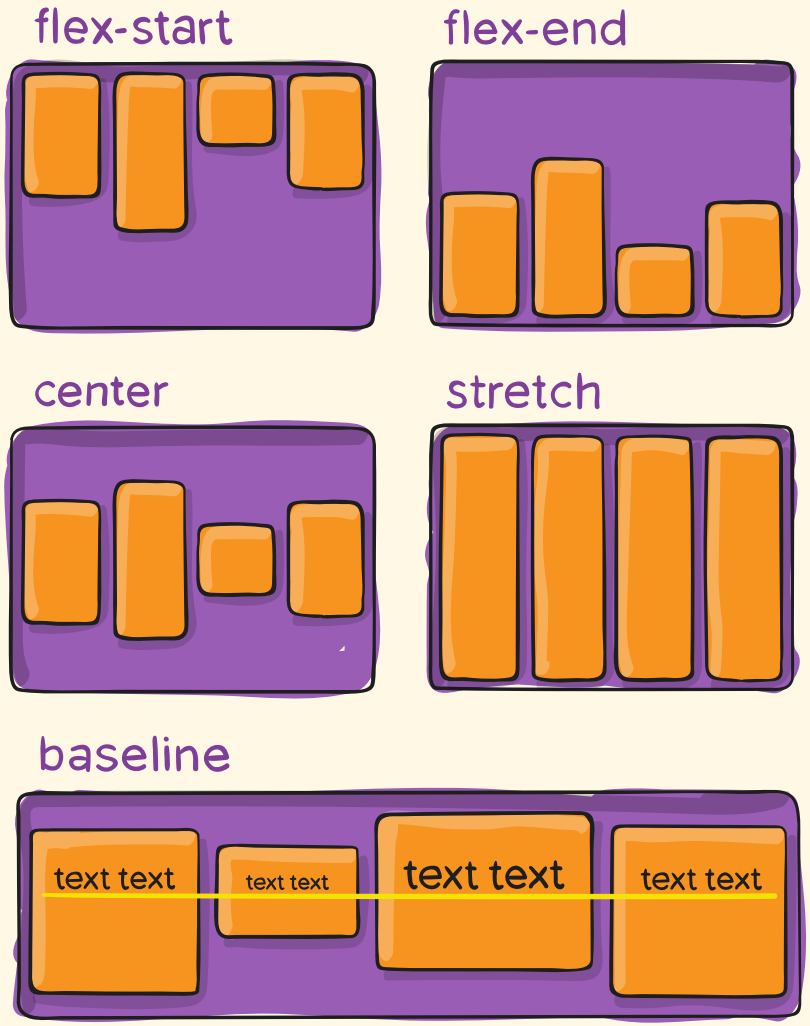
Container: `justify-content`

- `flex-start` (default), `flex-end`: items packed toward the start/end of the flex direction
- `start`, `end`: same as `flex-start`/`flex-end`, but wrt the writing-mode direction
- `left`, `right`, `center`: items are packed toward the left/right/center of the container
- `space-between`, `space-around`, `space-evenly`: items evenly distributed in the line



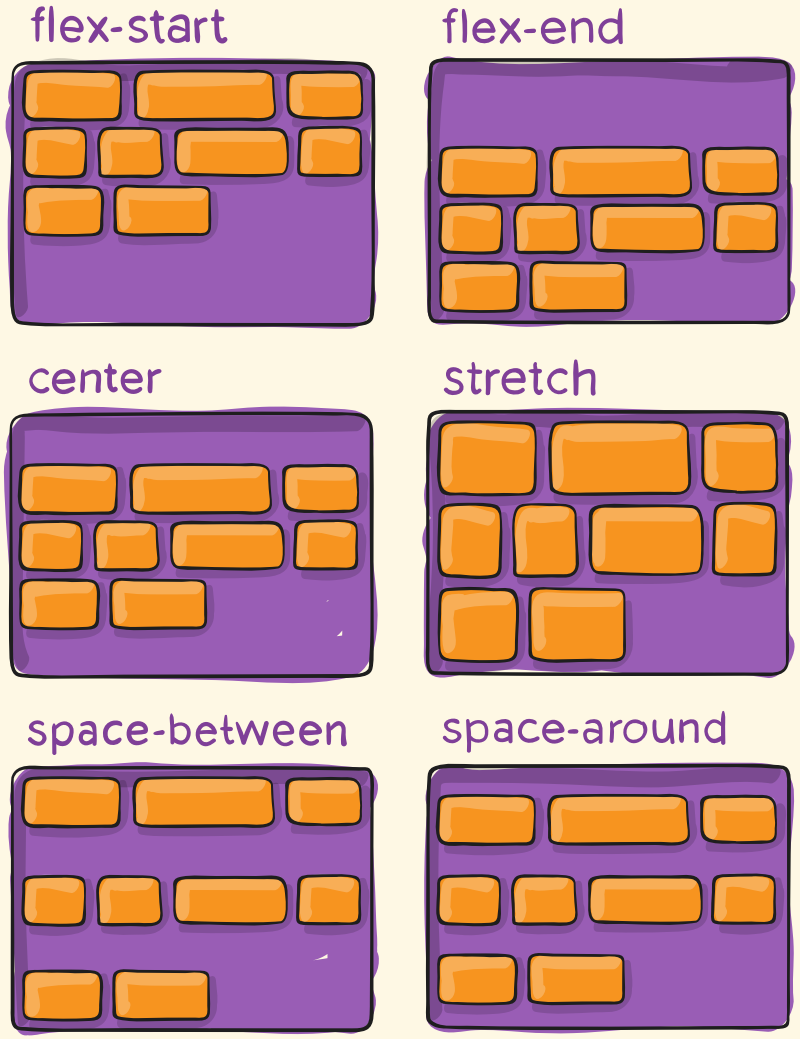
Container: `align-items`

- `stretch` (default): items stretch to fill the container (still respect `min-width`/`max-width`)
- `flex-start`, `start`, `self-start`: items placed at the start of the cross axis. The difference of them is about respecting the `flex-direction` or the writing-mode rules
- `flex-end`, `end`, `self-end`: items placed at the end of the cross axis
- `center`: items centered in the cross-axis
- `baseline`: items aligned such as their baselines align



Container: `align-content`

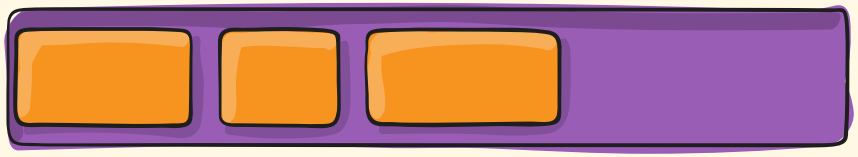
- `normal` (default): items packed in their default position
- `flex-start`, `start`: items packed to the start of the container, `flex-start` wrt the `flex-direction`, while start wrt the writing-mode direction
- `flex-end`, `end`: items packed to the end of the container
- `center`: items centered in the container
- `stretch`: lines stretch to take up the remaining space
- `space-between`, `space-around`, `space-evenly`: items evenly distributed



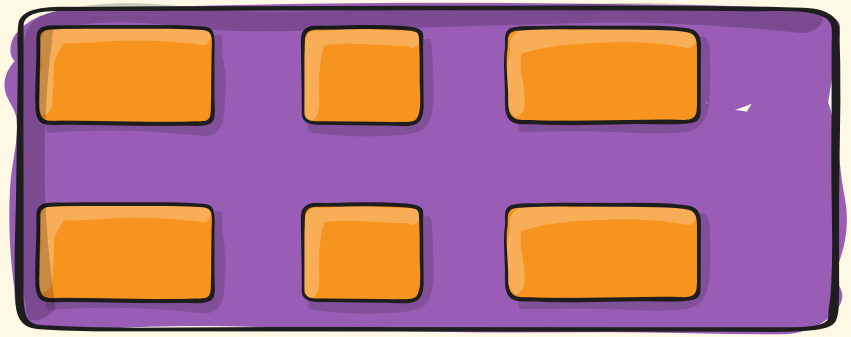
Container: gap, row-gap, column-gap

- To control the gap between rows and columns
- Examples:
  - `gap: 10px`
  - `gap: 5% 0.5rem`
  - `row-gap: 2em`
  - `column-gap: 10vh`

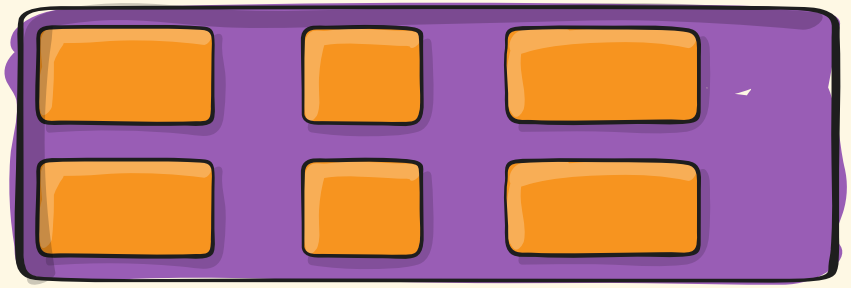
gap: 10px



gap: 30px

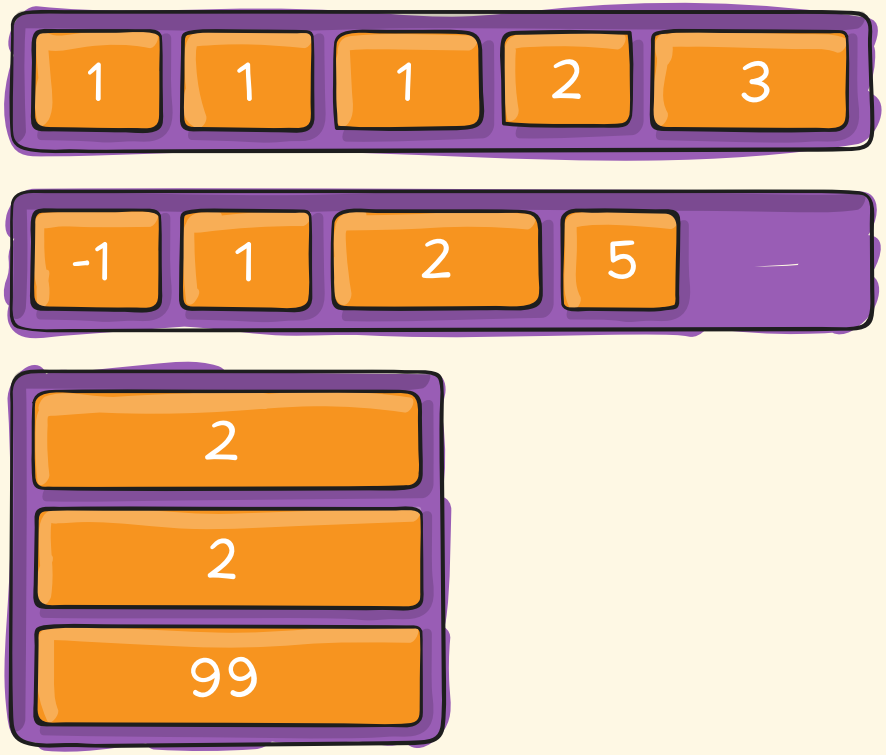


gap: 10px 30px



Items: `order`

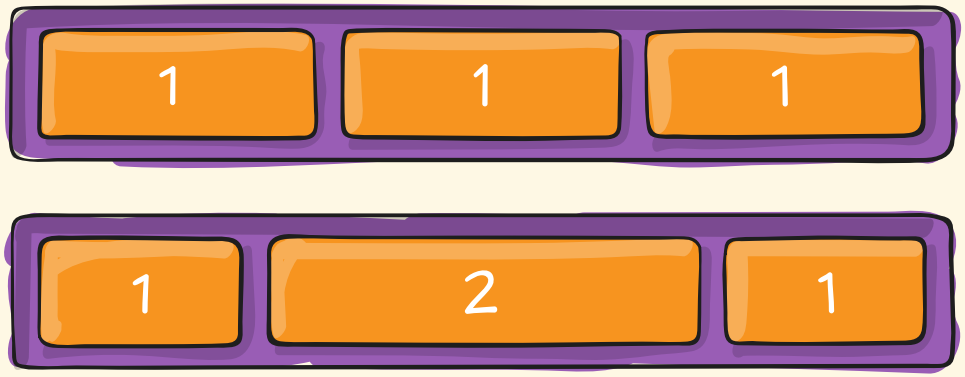
- Helps to alternate the default order
- Items with same value are laid out in their source order
- Default value: 0




```
.item {
   order: 5;
}
```

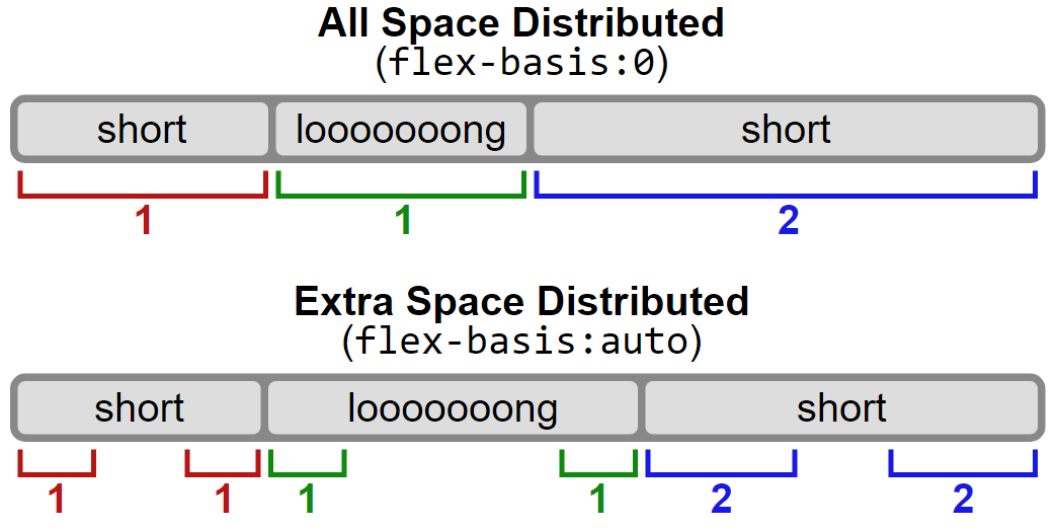
Items: `flex-grow`

- Defines the ability for an item to grow if necessary: how much the remaining space is distributed to each item
- Value serves as a proportion: an item set to 2 would take up twice as much as space as an item set to 1
- Default value: 0



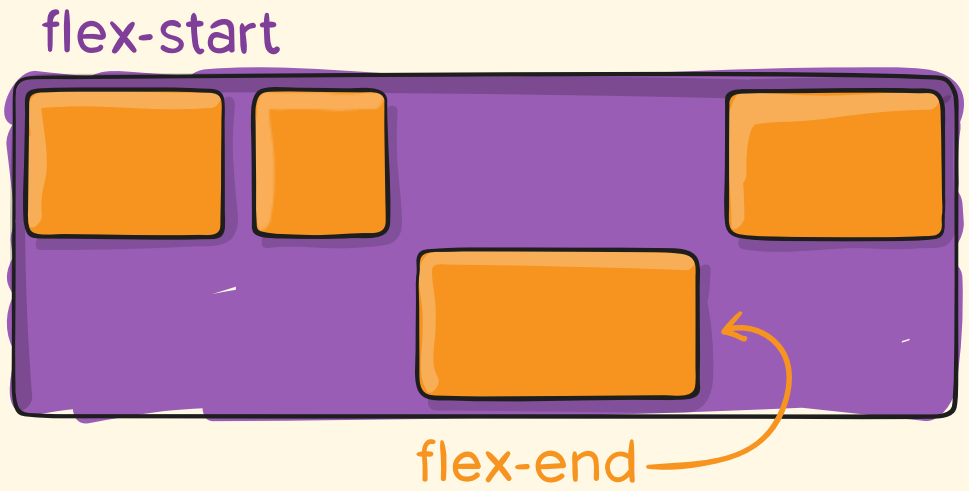
Items: `flex-basis`

- Defines the default size of an element before the remaining space is distributed
  - Can be a length: `20%`, `100px`, `10rem`
  - `0`: the extra space around content isn't factored in
  - `auto`: the extra space is distributed based on its `flex-grow` value



Items: `flex-self`

- Allows the default alignment (or the one specified by container's `align-items`) to be overridden for individual flex items
- See `align-items` for the values



Two-Value `display` Property

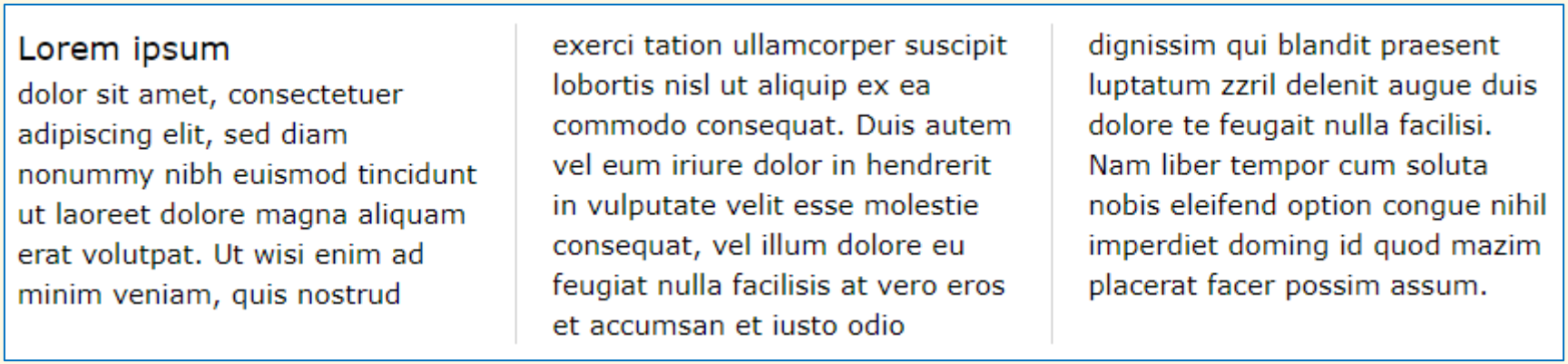
- Refactored into outer and inner types ➔ more generalized way
- Outer type:
  - `block`
  - `inline`
- Inner type:
  - `flow`
  - `flow-root`
  - `flex`
  - `grid`
  - `table`

Legacy (precomposed)	New (two values)
`block`	`block flow`
`flow-root`	`block flow-root`
`inline`	`inline flow`
`inline-block`	`inline flow-root`
`flex`	`block flex`
`inline-flex`	`inline flex`
`grid`	`block grid`
`inline-grid`	`inline grid`
`table`	`block table`
`inline-table`	`inline table`

Multi-column Layout

Introduction

- Allows to define multi-column layouts
- Basic example
  - ```
    .container {
      column-count: 3;
    }
    ```

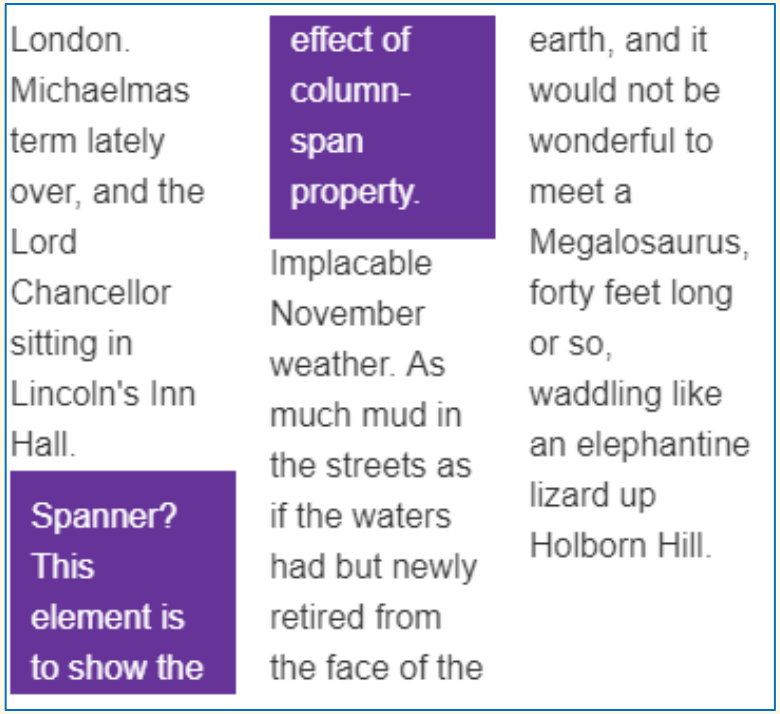

    ```

Layout Options

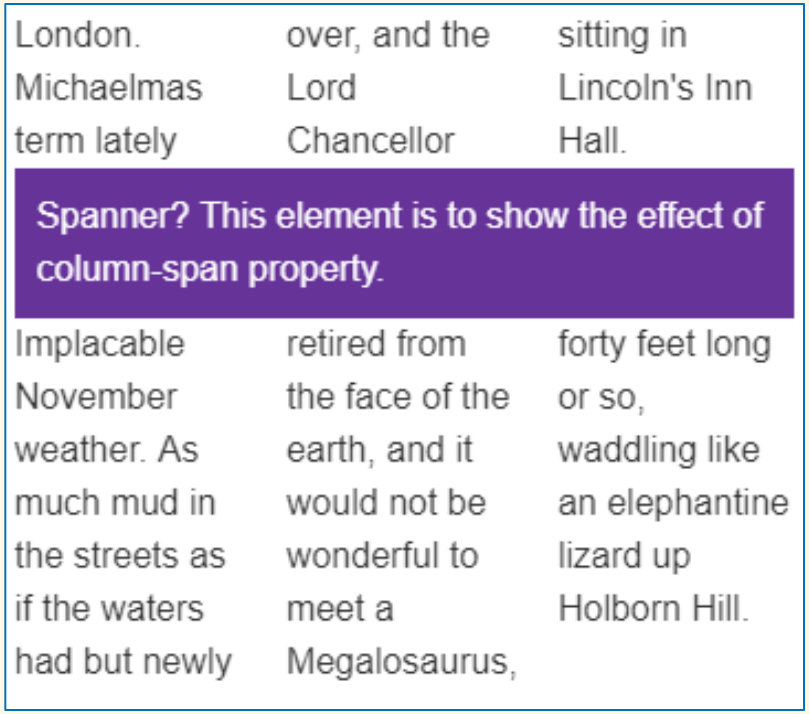
- Minimum width of columns:
  - `column-width: 100px;`
- Shorthand for `column-width` and `column-count`:
  - `columns: 100px 3;`
- Gap between columns
  - `column-gap: 40px;`
- Style of the rule between columns
  - `column-rule: 2px solid red;`
  - `column-rule-style: solid|dotted|dashed|...;`
  - `column-rule-color: red;`
  - `column-rule-width: 2px;`
- Column filling mode:
  - `column-fill: balance|auto;`

Element Span

- Number of columns an element should span across
  - `column-span: all|none;`



`column-span: none`



`column-span: all`

More Features

CSS Variables (aka. Custom Properties)

- Entities that contain specific values to be reused throughout a document
- Accessible using `var(…)`
- Case-sensitive

- Example:

```
element {
  --main-color: brown;
  background-color: var(--main-color);
}
```

Variables in `:root` pseudo-class

- In order to make the variables anywhere throughout the document

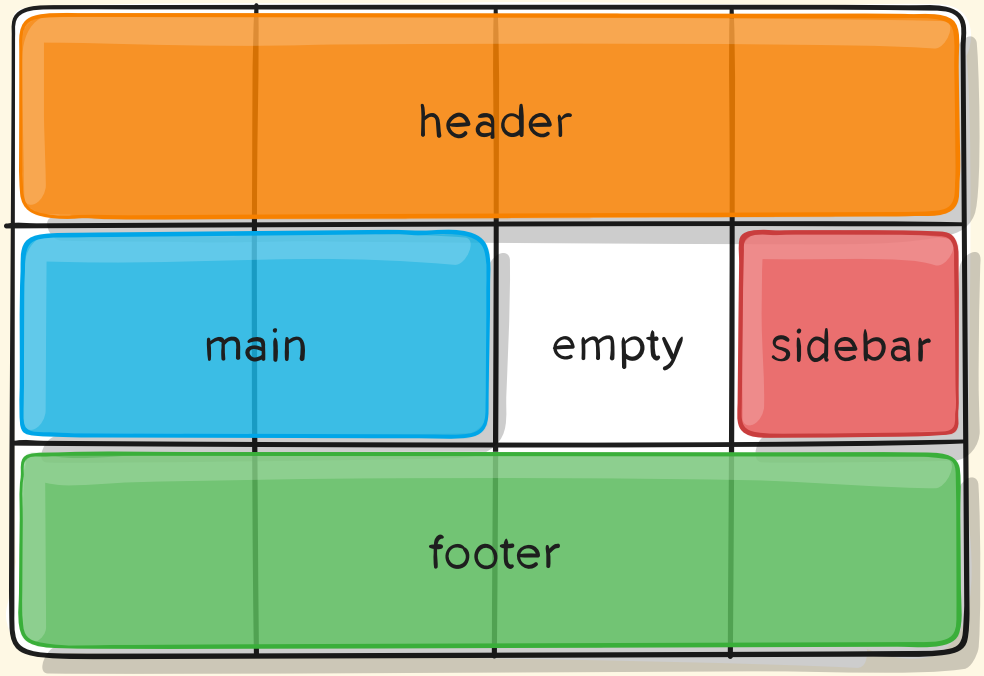
```
:root {
  --color: brown;
  --size: 10px;
}

.one {
  background-color: var(--color);
}

.two {
  --size-2: calc(var(--size) + 5px);
  width: calc(var(--size) * 2);
}
```

Grid Layout

- A 2D grid-based layout system that helps to design grid-based layouts
- To be self-studied



Even More

- Container queries
- CSS nesting
- Scroll driven animations
- View transitions
- …